IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 16 May 2025, accepted 29 May 2025, date of publication 17 June 2025, date of current version 30 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3580340

SURVEY

The Applications of Artificial Intelligence and Digital Twin in Power Systems: An In-Depth Review

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ABSTRACT The accelerated evolution of Artificial Intelligence (AI) technology has garnered widespread attention and acclaim across various domains, owing to its remarkable performance in diverse applications. Power systems supporting modern life are no exception to this transformative wave. This paper stands as the first comprehensive review of various AI techniques in power systems, spanning applications such as load forecasting, security assessment, voltage stability assessment, load shedding (LS), false data injection attack (FDIA) detection, state estimation and localization, fault detection and location, and power quality disturbances (PQDs) detection. In addressing the challenges inherent in the practical implementation of AI in power systems, this study introduces two potent tools: the strategic utilization of transfer learning (TL) in conjunction with AI algorithms and the leveraging of digital twin technology. The integration of these approaches substantially enhances the performance and accuracy of AI models. Overall, this work aims to contribute to the foundational knowledge necessary for harnessing the full spectrum of AI's capabilities and in advancing the shift toward a sustainable energy future.

INDEX TERMS Artificial intelligence, digital twin, power systems, load forecasting, security assessment, voltage stability, load shedding.

I. INTRODUCTION

In the last few decades, electric power systems have undergone a significant transformation. The traditional model of vertically integrated and regulated systems has given way to a modern electrical framework characterized by non-regulated markets, the integration of renewable energies, the deployment of smart grids, and increasing the penetration of distributed generations [1]. This transition has resulted in a growingly complex grid structure and increased non-linearity. The continuous increase in the share of dynamic loads and renewable energy sources within modern power grids has introduced new challenges that highly impact power system operation [2], [3]. Artificial intelligence (AI) algorithms are strategically employed in response to the profound transformations observed in electric power systems, marked by the shift from traditional, regulated models to a contemporary framework featuring non-regulated markets, renewable energy integration, smart grids, and distributed generation. These dynamic applications of AI optimize and enhance various facets of power system operation. Given the increasingly intricate grid structure characterized by heightened non-linearity and the challenges introduced by the growing proportion of dynamic loads and renewable energy sources, these algorithms play a pivotal role in tackling the complexities of the evolving grid landscape.

The associate editor coordinating the review of this manuscript and approving it for publication was Zhehan Yi.

A. LITERATURE REVIEW AND RESEARCH GAP

This paper is the most comprehensive review thus far, thoroughly exploring key areas where AI is harnessed in power

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systems, specifically in load forecasting, security assessment, voltage stability assessment, and load shedding (LS). While several studies have been conducted during the last years, they often encountered some drawbacks that limited their comprehensiveness and scope, including:

- The exclusive focus on Machine Learning (ML) or Deep Learning (DL) based techniques in the reviewed articles [4], [5], [6], [7], [8]: This paper goes beyond such limitations and comprehensively studies the application of various AI techniques by recognizing the broader spectrum of AI, including rule-based systems and metaheuristic optimization methods. This method positions the paper as the most complete AI-based review work in power systems to date.
- A narrow focus on reviewing only one application within the power system [4], [5], [8], [9], [10], [11], [12]: While prior studies focused solely on evaluating a single application within the power system, this paper diverges from this trend. It distinguishes itself by delivering an exhaustive review that spans the fundamental facets of power systems. These analyses include load forecasting, security assessment, voltage stability assessment, and LS. Moreover, the paper concisely explores other significant applications, namely, state estimation, false data injection attack (FDIA) detection and localization, fault detection and location, and detection and classification of power quality disturbances (PQDs). Through this inclusive approach, the paper solidifies its standing as a comprehensive reference in AI applications for power systems.
- Furthermore, this paper offers a comprehensive overview of research articles on digital twin technologies, emphasizing their application in power systems. Digital twin technologies emerge as a novel tool that addresses practical barriers to AI implementation in power systems. Notably, research on digital twin applications in power systems is still in its early developmental stages. Therefore, this work is notable for being the first to encompass and critically assess this emerging aspect of digital twin research, pioneeringly contributing to digital twin technologies in the context of power systems. Additionally, it is the preliminary study to simultaneously examine both digital twin and AI applications in power systems, thereby expanding the breadth of insights into the integration of these state-of-the-art technologies.

In the realm of AI applications in power systems, a meticulous analysis of three critical aspects of grid stability assessment is offered in [13], including voltage stability, transient stability, and frequency stability. This paper is a valuable resource for those seeking a comprehensive insight into the holistic assessment of power grid stability. However, it is crucial to acknowledge that while it excels in addressing stability assessment, it does not extend its coverage to other essential applications (e.g., load forecasting and FDIA within power systems). A similar focus is shared in [14] by concentrating solely on ML and fuzzy logic algorithms in grid stability assessment. Nevertheless, it is essential to highlight that it lacks consideration of metaheuristic optimization algorithms, leaving a gap in its coverage of AI methodologies within the broader landscape of power systems applications. Finally, [15] emerges as the most comprehensive review in the existing literature on AI applications in power systems, offering a meticulous examination of various applications within power system operation and monitoring. However, its primary focus is on DL algorithms, falling short in covering some essential applications, such as LS, within the broader landscape of AI applications in power systems since the current study stands out as the first to delve into references specifically addressing LS within the context of AI applications in power systems.

B. CONTRIBUTIONS AND ORGANIZATION OF THE PAPER

This research addresses limitations in previous studies by covering a wide range of AI approaches applied across various power system applications rather than focusing on a single aspect. While earlier work often targets either data-related or physical system challenges, we emphasize how integrating Transfer Learning (TL) and Digital Twin can effectively tackle both to offer a more practical solution. To the best of the authors' knowledge, this is the first review paper on the application of AI techniques in power systems, highlighting the following main contributions:

- A concise yet thorough review of AI and its subcategories. This section serves as a foundational exploration, laying the groundwork for a comprehensive understanding of their applications in the context of power systems
- Different power system applications are explored, including load forecasting, security assessment, voltage stability, LS, and a brief review of additional areas such as state estimation, FDIA, PQD, and fault detection, with each application accompanied by presenting a taxonomy of the most recent references pertinent to the specific domain
- Throughout this comprehensive investigation, diverse AI techniques, such as rule-based methodologies, ML techniques, and metaheuristic optimization methods, are considered for different applications.
- Examining challenges in practical implementation of AI in power systems, this study introduces two pivotal tools—TL and digital twin technology.

Fig. 1 provides a detailed visual summary, outlining the primary domains and applications that will be thoroughly examined in this study. The subsequent sections of the paper are structured in the following manner:

- Section II is a concise overview of the primary categories of AI approaches and their respective subcategories to provide a foundational understanding of each methodology.
- Section III: provides an in-depth examination of how various AI methodologies are applied to enhance power

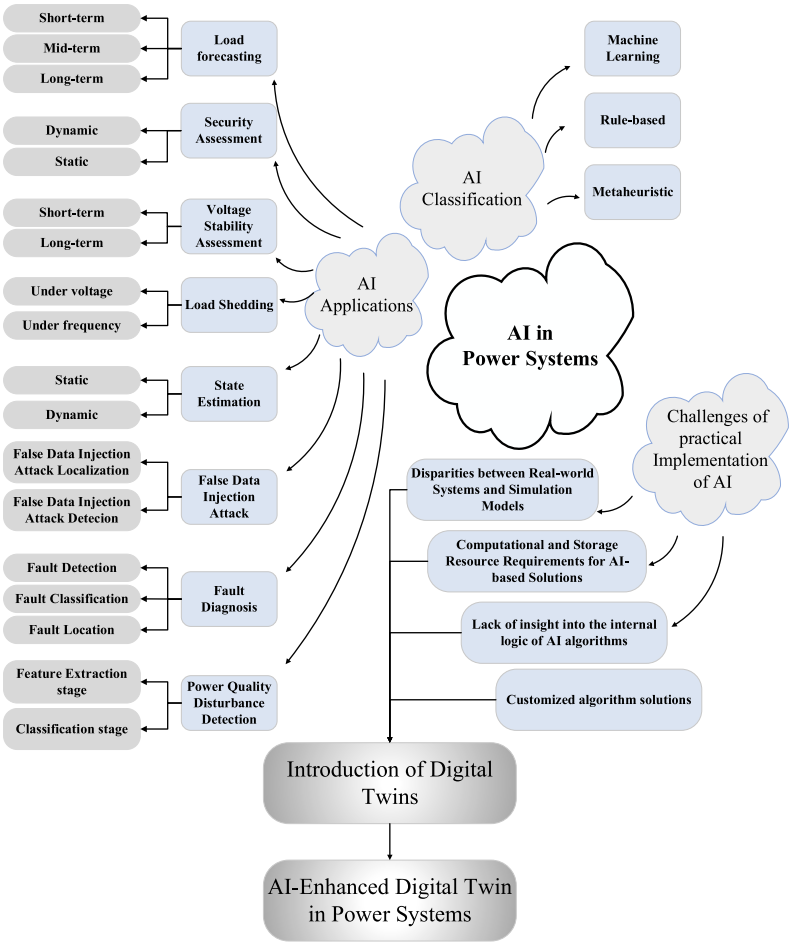


FIGURE 1. Overview of key domains and applications explored in this study.

system operations. Key applications discussed include load forecasting, security assessment, voltage stability assessment, load shedding, and other critical areas.

- Section IV: Exploring challenges in the practical implementation of AI techniques in power system applications and introducing Digital Twin to overcome these barriers through its integration with AI (especially TL).
- Section V: A case study of TL Implementation in power system applications and a comparative analysis with traditional DL methods.
- Section VI: Conclusion and key takeaways.

II. BREAKING DOWN AI APPROACHES: MACHINE LEARNING, RULE-BASED AND METAHEURISTICS

AI encompasses a broad domain of research that is focused on advancing algorithms and systems that empower computational devices to acquire knowledge, exercise informed decision-making, and perform tasks conventionally linked to human intelligence. These tasks encompass various cognitive abilities, such as problem-solving, learning from experience, comprehending natural language, and recognizing complex patterns. The overarching objective of AI is to endow

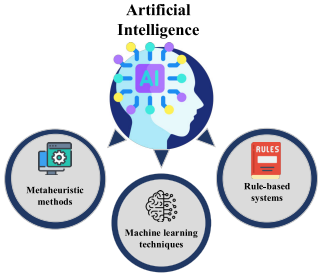


FIGURE 2. Categorization of artificial intelligence.

machines with the capacity to exhibit intelligent behavior and execute tasks that have traditionally been attributed to human capabilities.

AI systems can be broadly classified into three primary paradigms: rule-based systems, ML techniques, and metaheuristic methods, as presented in Fig. 2 [18]. Further elucidation on these paradigms and their applications will be presented in this discussion.

A. MACHINE LEARNING TECHNIQUES

ML is a set of methodologies widely employed in contemporary research endeavors. Here, we will revisit fundamental concepts, categorized into four subgroups of supervised,

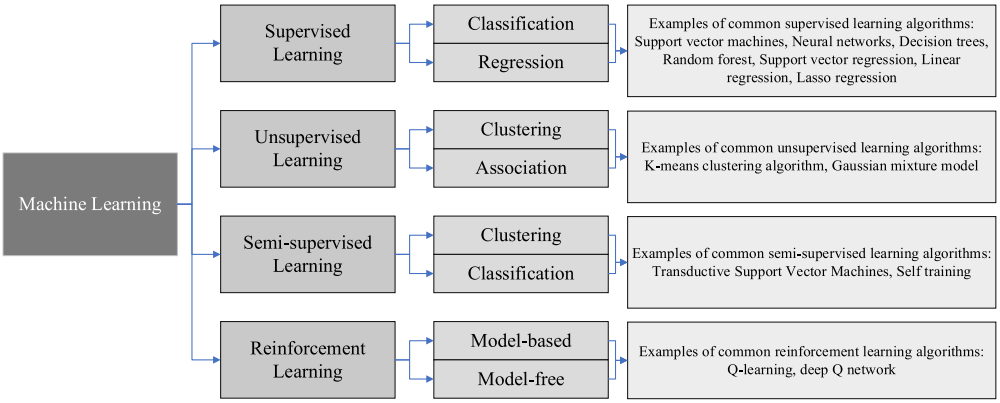


FIGURE 3. Machine learning algorithms' classification [16], [17].

semi-supervised, unsupervised, and reinforcement learning (RL), according to Fig. 3 [16], [17]. The initial category of ML is known as supervised learning, wherein the learning process operates under the guidance and utilization of labeled datasets. Labeled datasets consist of data instances that are accompanied by known target answers or output labels.

Unlike supervised learning, unsupervised learning involves training ML models exclusively on input features without associated labels. The main objective of unsupervised learning is to extract meaningful patterns, structures, or insights from the data without the explicit guidance of human-labeled annotations. Unsupervised learning techniques encompass various tasks, including methods for reducing dimensionality such as principal component analysis (PCA), clustering algorithms like k-means clustering, generative modeling methods such as autoencoders (AEs), and anomaly detection approaches [19].

By exploring the inherent structure and relationships within the data, unsupervised learning enables the models to uncover hidden patterns or groupings autonomously, reducing the dependency on human intervention or labeled information.

To mitigate constraints linked to the scarcity of labeled data in supervised learning, the emergence of semi-supervised learning techniques has provided a viable solution. By incorporating both labeled and unlabeled data during the training process, semi-supervised learning approaches aim to enhance classification performance. Alternative ML algorithms have undergone both theoretical and experimental validation, demonstrating promising performance and rapid learning speeds in the context of supervised classification tasks. However, these algorithms may not perform satisfactorily when confronted with imbalanced binary classification tasks, often demonstrating a bias towards the majority class. Moreover, considering the scarcity of labeled training data in real-world scenarios, there exists an imperative need to develop an efficient semi-supervised adaptation of these algorithms tailored explicitly for imbalanced binary classification tasks [20]. Moreover, RL is the last learning technique that warrants mention in this context. RL entails an agent's observation of the environment's state and subsequent selection of actions to accomplish a task defined by a pre-established reward signal. This reward signal serves as a teaching mechanism for the agent, guiding it to perform actions that maximize the expected future rewards. Through this process, the agent leverages past experiences to enhance performance in the given task. RL algorithms have demonstrated notable achievements in diverse applications, particularly in complex robotic control domains [21].

B. METAHEURISTIC METHODS

Metaheuristic optimization techniques encompass a group of algorithms used to solve optimization problems, including the search for optimal hyperparameters in models and controllers [18]. These methodologies can be classified into trajectory-based and population-based methods. Trajectory-based metaheuristic methods refine and enhance individual candidate solutions through iterative modifications, whereas population-based metaheuristic methods maintain and improve multiple candidate solutions simultaneously [22]. Prominent examples of trajectory-based and population-based methods are represented in Fig. 4 [23], [24]. These diverse approaches collectively contribute to the field of metaheuristic optimization, providing flexible and robust techniques for solving complex optimization problems. Population-based metaheuristics frequently draw inspiration from natural phenomena, leading to bio-inspired approaches to solving optimization problems. These bio-inspired population-based metaheuristics can be broadly categorized into two main classes: evolutionary algorithms and swarm intelligence algorithms. Genetic Algorithms (GA) and Differential Evolution are widely recognized and popular algorithms in the first category. In the second category, Ant Colony Optimization and Particle Swarm Optimization (PSO) are well-known examples [25]. Single-solution metaheuristics, on the one hand, primarily rely on local search procedures, aiming to incorporate intelligent mechanisms that effectively exploit local search moves. Tabu

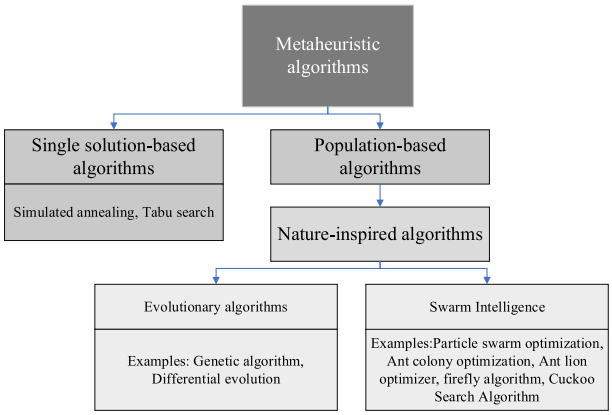


FIGURE 4. Classification of metaheuristic algorithms used in solving optimization problems [23], [24].

Search, Simulated Annealing, Iterated Local Search, and Greedy Randomized Adaptive Search Procedure are distinct single-solution metaheuristics with unique characteristics. These metaheuristics offer valuable strategies for addressing combinatorial optimization problems.

C. RULE-BASED SYSTEM

Rule-based systems represent a distinctive category of AI methodologies that facilitate the assimilation of human expertise into computational systems. By encoding human expertise in rules, rule-based systems can make informed decisions and solve complex problems that reflect human reasoning [26]. Due to their intended emulation of human experts' decision-making abilities, Rule-based systems are commonly denoted as expert systems. Based on how they represent and manage knowledge, rule-based systems can be classified into fuzzy logic rule-based systems and Boolean logic rule-based systems, as depicted in Fig. 5 [18].

Boolean logic systems, also called classical or crisp logic systems, constitute a prominent category of rule-based systems wherein the foundational logic relies on binary values, typically denoted as true (1) or false (0). Such systems are designed to operate with precise or deterministic data, adhering to the fundamental principles of classical logic [27]. Nevertheless, conventional Boolean logic alone may prove inadequate in resolving some intricate issues [28]. In contrast, a fuzzy rule-based system represents an alternative rule-based AI technique that adopts fuzzy sets and fuzzy logic to represent knowledge effectively and cope with uncertainty. Fuzzy logic finds its rationale in how human cognition handles imprecise information to make judgments. A fuzzy rule-based system encompasses crucial components, including a knowledge base comprising a dataset and a rule base, a fuzzy inference engine, and fuzzification and defuzzification modules. The knowledge base comprises a dataset holding domain-specific information and a rule base with fuzzy if-then rules encoding human expertise and knowledge. The fuzzy inference engine processes these rules to draw conclusions and make decisions based on input data.

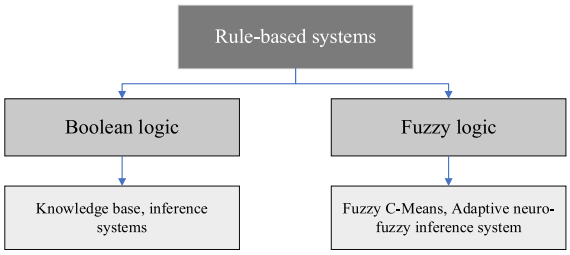


FIGURE 5. Classification of rule-based methods [18].

Fuzzification converts crisp inputs into fuzzy sets, allowing the handling of uncertainty. After producing fuzzy output sets, defuzzification aggregates them using techniques like centroid defuzzification to obtain precise results. Fuzzy rule-based systems are advantageous in dealing with systems that involve imprecision, uncertainty, or ambiguity, making them suitable for applications such as control systems, decision-making processes, and pattern recognition tasks [26], [29]. Transitioning from the exploration of various AI categories, the subsequent section delves into a comprehensive examination of diverse applications of AI within power systems.

III. COMPREHENSIVE REVIEW OF ARTIFICIAL INTELLIGENCE APPLICATIONS IN POWER SYSTEM

This segment explores the advanced application of AI techniques, including ML, metaheuristic optimization algorithms, and fuzzy logic, within the realm of power system operation and control. This integration enhances various aspects, from load forecasting to power system stability, marking a new era of efficiency and adaptability in electrical engineering.

A. LOAD FORECASTING

Precise load forecasting assumes paramount importance in effective power system management and strategic generator planning, particularly during periods of rapid escalation in power consumption and inadequacy in the electric power reserve rate [30].

The difficulty in obtaining precise load forecasting results lies in the intricate nature of grid management. Numerous elements, such as consumer behavior, appliance types, usage differences, the growing presence of renewable energy sources, and volatile pricing, add to this complexity [31]. Consequently, achieving precise load forecasts remains a formidable undertaking, essential for providing reliable and economically sustainable electric energy to consumers while ensuring the security of power supply [32]. Obtaining accurate predictions regarding network load behavior and changes in the short, mid, and long term is imperative for daily electrical network operators and strategic planners responsible for national power grids in any country [33].

1) LOAD FORECASTING CATEGORIZATION

Categorically, load forecasting can be divided into three main distinct types: short-term load forecasting (STLF), mid-term

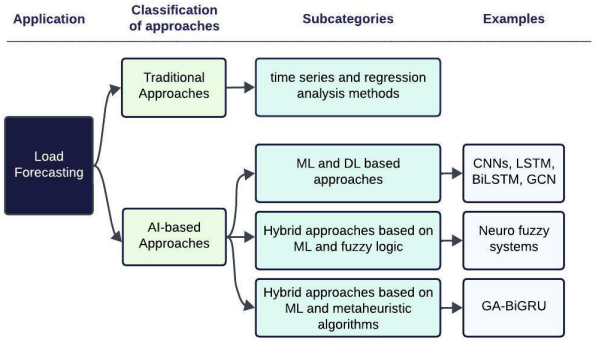


FIGURE 6. **Classification of load forecasting approaches explored in this paper.**

load forecasting (MTLF), and long-term load forecasting (LTLF).

STLF and MTLF load forecasting methodologies find application in power system operations, serving operational purposes with distinct timeframes. Typically, STLF and MTLF are employed for hour-ahead, day-ahead, month-ahead, or year-ahead forecasting, respectively. In contrast, LTLF assumes a role in facilitating power system infrastructure planning, extending its horizon beyond a year to encompass longer periods [34]. Each of these forecasting methods is subject to a multitude of influencing factors. STLF, in particular, assumes a fundamental role as it serves as the foundation for the start-up, shut-down, dispatch, and operation planning of the power grid internal units [35]. It is an essential input for power system operators, enabling real-time decision-making in electricity supply and demand management. Due to their impact on heating and cooling requirements, short-term load patterns are directly influenced by weather conditions, including temperature, humidity, and wind speed. Moreover, special events, holidays, and economic indicators, such as industrial activities and business hours, significantly shape short-term electricity consumption patterns [36].

2) ARTIFICIAL INTELLIGENCE-BASED APPROACHES FOR LOAD FORECASTING

Numerous load forecasting methodologies have been devised, categorized into traditional and AI-based approaches. Fig. 6 represents the classification of various approaches for load forecasting in this paper.

The traditional methods primarily encompass time series and regression analysis methods and predominantly rely on linear models, which may prove inadequate to address the inherent nonlinearity prevalent in load data, and also have limited modeling ability for external factors such as weather changes [37]. However, the scope of this paper does not encompass traditional methods, as it focuses explicitly on exploring AI-based methods for load forecasting.

In the context of exploring related works, different algorithms for STLF have been studied so far. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks are prominent methodologies in DL, increasingly used for power load forecasting. CNNs excel at extracting deep features and filtering noise, while LSTMs effectively capture sequence patterns and overcome issues of gradient vanishing or explosion found in traditional RNNs [38]. In [36], the authors introduce a STLF hybrid DL model by integrating CNN and LSTM. It exhibits a unique architecture with separate pathways for CNN feature extraction and LSTM for capturing long-term dependencies in the input data. Similarly, in [35], a STLF model based on Bi-directional LSTM (BiLSTM) is proposed. BiLSTM is a variant of the LSTM network that combines forward and backward LSTM to capture bi-directional dependency in the input data, allowing the model to process long-term sequential information effectively. The proposed BiLSTM-based model leverages historical load data from a regional power grid and four industries to enhance STLF accuracy and performance. Another hybrid DL model for STLF is addressed in [39], which utilizes graph convolutional network (GCN) and LSTM to extract spatial and temporal characteristics from massive data for STLF. The model achieves enhanced load prediction accuracy and training efficiency by utilizing multi-dimensional data, including meteorological, date, and regional factors.

Hybrid models based on ML and fuzzy logic leverage the respective advantages of both approaches to tackle complex and uncertain problems. Fuzzy logic enables the representation and processing of linguistic information, while ML techniques provide robust pattern recognition and data-driven learning capabilities. One prominent example of such a hybrid model is the neuro-fuzzy system, which combines neural networks and fuzzy logic to effectively handle complex and uncertain data by harnessing the learning capabilities of neural networks and the interpretability of fuzzy logic's linguistic representation. To this end, a hybrid approach called WLANFIS for STLF is proposed in [40], combining weighted least square, neural networks, and adaptive neuro-fuzzy inference system (ANFIS). Likewise, a hybrid STLF method based on the fuzzy combination weights and empirical mode decomposition process (FCW-EMD) alongside the Bat algorithm-optimized support vector machine (SVM) and the Kalman filtering process (KF-BA-SVM) is presented in [41]. Unlike existing methods based on a single model, this approach integrates crucial processes and introduces two new theories: FCW for similar days selection and BA-SVM for SVM parameter optimization.

Among STLF hybrid metaheuristic DL models, authors in [42] propose an integrated GA-bidirectional gated recurrent unit (Bi-GRU) hybrid data-driven technique (GA-BiGRU) for STLF. The proposed hybrid evolutionary DL approach has the advantage of using BiGRU with an attention mechanism that allows access to both past and future data, enabling better load prediction accuracy. GA is utilized to find optimal or near-optimal parameters for DL models, which improves their performance by accurate hyperparameter selection.

On the other hand, MTLF is crucial for maintaining a reliable power system in a smart city as it allows us to generate electricity according to future energy demands. Nevertheless, energy consumption exhibits dynamic fluctuations influenced by diverse factors, including governmental policies, weather variations, demographic changes encompassing population shifts, urbanization and migration, and economic indicators, such as gross domestic product growth and industrial activity. Therefore, meticulous consideration of diverse determinants becomes indispensable in the context of MTLF [43]. For instance, a MTLF model considering policy factors in China's evolving power load development is proposed in [44]. The model integrates a hierarchical policy-influencing factor index system to systematically depict the influence of economic factors and policies on load. The load prediction under the influence of policy factors is accomplished by applying the weighted fuzzy cluster analysis method. The proposed model effectively addresses the challenges of load forecasting under policy changes and outperforms traditional methods in terms of prediction accuracy and practicality.

Finally, within the scope of LTLF, electricity consumption forecasts of the forthcoming years serve as principal inputs in the grid planning department, specifically guiding grid renovation and expansion endeavors [38]. These forecasts play a pivotal role in shaping the strategic decisions related to augmenting and modernizing the power grid infrastructure, ensuring its resilience and adaptability to meet evolving electricity demands. The study in [45] examines contemporary approaches employed in predicting long-term hourly electricity demand at a collective level, encompassing regional or national scopes, for projections spanning a period of 10 to 50 years. They analyze the intricate obstacles inherent in these methodologies within the context of an anticipated future energy landscape characterized by heightened integration of renewable energy sources and a more intricate interrelation between the power industry and the domains of construction and transportation. Additional sources investigating load forecasting methods are listed in Table 1.

The effectiveness of AI-based forecasting models is well-established, yet their performance often relies on extensive datasets for training. In the case of long or midterm forecasting, where data points are generated less frequently, obtaining enough data for model construction can be challenging. To address this issue, [43] propose a novel TL-based approach for monthly electric load forecasting in urban settings. In certain real-world situations, such as new constructions, infrastructure changes, insufficient data collection systems, data privacy regulations, and remote areas, TL can address data scarcity in load forecasting problems. Several studies [46], [47], [48], have demonstrated successful applications of TL in load forecasting, particularly in addressing data limitations in STLF and LTLF.

B. SECURITY ASSESSMENT

The online security assessment of power systems holds paramount importance in enhancing situational awareness and expediting emergency decision-making processes undertaken by operators. This critical endeavor catalyzes enabling operators to institute proactive regulatory measures and implement necessary corrections when confronted with disruptive contingencies [49]. The primary aim of this endeavor is to ascertain the prevailing security status of the power system, discerning whether it resides within a secure or insecure state. A secure state denotes a condition wherein no breaches of operational thresholds are anticipated under the current operating parameters, even in the event of unforeseen disturbances and contingencies. Furthermore, in a secure state, the power system successfully meets the required load demand. Conversely, an insecure state signifies that certain operational limits may be transgressed and that the load demand cannot be entirely or partially fulfilled [50].

1) CATEGORIZATION OF SECURITY ASSESSMENT METHODS

The concept of security assessment can be generally delineated into two principal classifications: dynamic security assessment (DSA) and static security assessment (SSA). As stated in [51], scrutinizing a power system's ability to endure and persist through a finite array of contingencies until reaching an acceptable steady-state state is formally referred to as DSA. The principal DSA classifications, as elucidated, encompass transient stability, voltage stability, and frequency stability. Managing power networks and solving associated problems require complex modeling and calculations, which can sometimes become overwhelming and disrupt decision-making [52]. Consequently, AI methods have been utilized to reduce the computational workload and enhance problem-solving efficiency. Nonetheless, these approaches have not substantially reduced the necessity for simulations, as authentic data is indispensable for diverse learning methodologies and testing outcomes. This data must originate from the control center's database or be generated through simulation. Security assessment necessitates a meticulous evaluation of many scenarios, encompassing load patterns, power generation, and system configuration. This becomes paramount considering the escalating integration of renewable energy sources. However, the realm of DSA encounters formidable hurdles in the preliminary generation of imperative datasets before implementing algorithms. Accordingly, [53] addresses a core challenge encountered by the majority of DSA methods prior to algorithm deployment: the generation of requisite datasets crucial for the advancement of dynamic security classification strategies. The paper introduces a computationally efficient and exceptionally scalable approach to generating the necessary datasets for training or testing DSA methodologies. They claim that training ML algorithms using the dataset generated by their approach outperforms other approaches when computation time is the same.

2) SECURITY INDICES AND MONITORING

DSA indices play a crucial role in the continuous evaluation of the real-time or near-real-time stability and security of

TABLE 1. Exploring AI-based load forecasting methods in recent papers.

Ref	Forecast horizon	Input data	Dataset	Prediction method	Forecasted variable	Country/ year of study
[54]	Short-term	Electric load data, temporal factors, weather data	Load datasets of a region in Shanxi Province, 2014	Conditional generative adversarial network (CGAN) with semi-supervised regression and CNN	Daily load	China - 2023
[38]	Short-term and mid-term	Electric load data at 15-minute intervals plus weather data: temperature, wind speed and direction at daily intervals	10th Teddy Cup Data Mining Challenge	Multi-task learning, CNN, and LSTM	Electricity load for the next 10 days	China - 2023
[55]	Short-term	Hourly electric load data	IEEE dataset's hourly power consumption data	Deep neural networks	Hourly load	China-2024
[44]	Mid-term and long-term	Historical electricity consumption data	Electricity consumption data of the whole society of a province in China spanning from 2008 to 2020	Weighted Fuzzy Cluster Analysis	Annual load	China,2021
[39]	Short-term	Temperature, dew point temperature, humidity and precipitation	Public data from the PJM official website in the United States spanning from March 2003 to April 2017	GCN and LSTM	Daily load	China, 2023
[56]	Short-term	Hourly electric load data	electrical load time-series data from 81 power substations in Tehran at 1-hour time steps for 42 months from July 2015 to January 2019	SAC-ConvLSTM algorithm, which combines Spatial Auto Correlation with Convolutional Long Short-Term Memory networks	Future power consumption loads over a two-week horizon.	Iran - 2024
[33]	Long-term	Daily maximum and minimum temperature and solar exposure, maximum daily load	Load data for 12 years, from 2005 to 2016, for the state of South Australia	DL model based on Stacked LSTM Network and GA	Annual peak load	Saudi Arabia, 2021
[57]	Mid-term	Temperature, relative humidity, wind speed, and solar radiation	Load data for 12 years, from 2005 to 2016, for the state of South Australia	Artificial neuro fuzzy inference system	Consumption of load on a weekly or monthly basis for 6 months	Nigeria, 2023
[48]	short-term	Hourly residential electric load data	Load data from the University of Nigeria Nsukka	Artificial neuro fuzzy inference system	Consumption of load on a weekly or monthly basis for 6 months	Nigeria, 2023
[58]	Mid-term and long-term	Historical electricity consumption data	China's electricity consumption data from 2000 to 2019	Improved Sparrow Search Algorithm for SVM which is termed as ISSA-SVM	Electricity consumption from 2014 to 2019.	China, 2022
[59]	Short-term	Historical electric load data	FE grid dataset and Dayton grid dataset	Hybrid Deep Learning and Beluga Whale Optimization (LFS-HDLBWO) method. The hybrid model they use is Convolutional Bidirectional Long Short-Term Memory with Autoencoder	Different hourly time horizons (from 1 hour to 24 hours ahead)	Saudi Arabia, 2024
[35]	Short-term and mid-term	Electric load data with 15min intervals and maximum and minimum daily load data for four industries	Historical load data of a regional power grid and four industries	BiLSTM for STLF and a hybrid DL model based on BiLSTM and random forest (RF) for MTLF	Daily load, which is being forecasted for the next three months.	China,2023

power systems. Among these indices, the transient stability margin (TSM) stands out as a widely employed metric, as it effectively indicates the transient stability level within a power system [60]. In addition to TSM, other notable

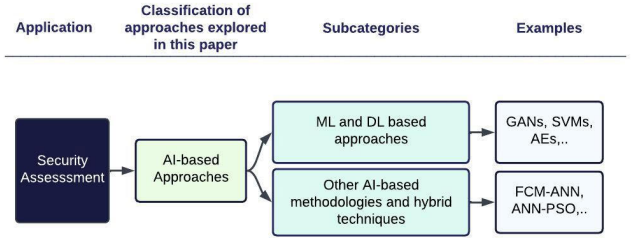


FIGURE 7. Classification of security assessment approaches explored in this paper.

indices in DSA include the transient stability index (TSI) and short-term voltage security, each contributing valuable insights to the ongoing monitoring and management of power system stability [61]. These indices collectively enable operators and engineers to make informed decisions to preemptively address potential instabilities and ensure the reliable operation of the electrical grid.

3) ARTIFICIAL INTELLIGENCE-BASED APPROACHES FOR SECURITY ASSESSMENT OF POWER SYSTEMS

Fig. 7 presents a categorization of AI-based approaches explored in this paper for security assessment. Data-driven AI methodologies, including but not limited to fuzzy inference systems [62], decision trees [63], [64], Gaussian Processes (GPs) [49], GANs [65], ANNs [66], SVMs [67] and others, have found extensive application in mitigating the constraints associated with traditional model-driven simulation methodologies within the domain of DSA. Supplementary exemplars, accompanied by comprehensive descriptions, are presented in Table 2.

In the context of power system analysis, there is a common assumption that the distribution of training data should align with that of real-time measurements for precise modeling. Nonetheless, this presumption might only sometimes be valid, particularly in the context of unforeseen system shifts or unencountered faults that diverge from the training data. Meeting the demand for equipping prevailing models to manage such unexpected scenarios effectively becomes crucial, especially considering the substantial multitude of potential faults inherent in practical power systems. To resolve this, [68] introduces a methodology that employs a single trained DSA model to evaluate unlearned faults, those not previously included in the training. Leveraging TL principles, the DSA model initially trained for a specific fault is adapted for a related but distinct fault. By minimizing distribution disparities between trained and unknown data, this approach significantly amplifies the extensibility of DSA models. Consequently, a limited set of DSA models can effectively cover many fault scenarios. Additionally, this transfer-based approach enables the identification of correlations between different faults, facilitating their aggregation.

In contrast, SSA examines whether buses' voltage and transmission line loading exceed their steady state limits following a disturbance, with the presumption that the system can ultimately achieve a stable equilibrium. A system is deemed statically secure if there are no limit violations during this stable state [50]. Numerous security indices within the academic literature are utilized to gauge the extent of exceeding desired limits. These metrics quantify the deviation of variables from their constraints during various contingency scenarios, offering insights into the severity of potential contingencies and their impacts on the system's state.

In the realm of system operation, the continuous monitoring of line loading and bus voltage stands as imperative requisites [69]. The performance index composite (PI_c) expounded in [70] offers a comprehensive evaluation of the system's performance by concurrently addressing the overloading of transmission lines and deviations in bus voltages. The study in [71] employs an additional index known as the Performance Index of Power (PI_p) and the performance index of voltage (PI_v) for security assessment. This approach entails an evaluation of the power index with a focus on active power flow and the priority of transmission lines.

SSA of the power system can be classified into two main components: system monitoring and contingency analysis [72]. The latter, comprising contingency screening and ranking, holds greater significance in ensuring system stability. In standard practice, grids undergo meticulous planning and operation, adhering to the N-1 security criterion, which guarantees system stability even in the face of plausible individual disruptions. A groundbreaking approach outlined in [73] introduces an innovative data-driven methodology explicitly crafted to expedite the N-1 contingency screening process, particularly when navigating through diverse and uncertain scenarios. The deep convolutional neural network (DCNN) serves as a regression tool for computing AC power flow. Once it undergoes thorough training, it is anticipated to exhibit strong generalization capabilities for unfamiliar power flow scenarios, resulting in a computational efficiency improvement of more than 100 times when compared to traditional model-based methods in N-1 contingency screening. In [74], a data-driven approach for fast cascading outage screening is introduced, grounded in DCNN and depth-first search (DFS). The DCNN is formulated as a regression tool to predict AC optimal power flow outcomes across various contingencies and assess the system security index.

The DFS algorithm scans the scenario tree, identifying the most severe cascading outage path using the security index value estimated by the DCNN.

The studies in [11], [51], and [93] also serve as a valuable resource, aiding the comprehensive understanding of power system security methods, encompassing static and dynamic assessment approaches. They contribute to a well-rounded comprehension of the field's methodologies and applications. Table 3 provides a summarized overview of diverse ML-based and hybrid security assessment techniques.

C. VOLTAGE STABILITY

Voltage stability, an integral facet crucial to the effective operation of power systems, represents the third key area

TABLE 2. AI-based security assessment methods in recent papers.

Ref	Security type	Input data	method	Summary of works' methodology - method	Test system and accuracy	year of study
[75]	Dynamic	The operational parameters of the grid (e.g., power generation/load and bus voltage magnitude)	ML-TL	The integrated TL method uses adversarial training to extract common features from various fault data, aligning source and target domains, enabling the application of a trained source domain classifier to unlabeled data in the target domain	New England 10-machine 39-bus system - 97.68%	2021
[68]	Dynamic	The operational variables of the system (e.g., power generation/load, and bus voltage magnitude.	TL	The method is based on TL, using labeled data from one source to train a model and applying it to unlabeled data in another target while considering differences in data space, distribution, and conditional probabilities between the two domains	New England 10-machine 39-bus system - 97.27%	2020
[76]	Dynamic	real-time operating variables from PMUs.	SMNB -CCF	The study uses Sparse Multinomial Naive Bayes (SMNB) for feature selection and Canonical Correlation Forest (CCF) for modeling, both of which are supervised ML techniques. The scheme utilizes real-time data from PMUs to predict security outcomes while addressing topology changes and data inconsistencies	IEEE 39-bus system and a practical 1648-bus system using PSS/E- 0.9813% & 0.9739%	2024
[60]	Dynamic	Operational variables of the system (e.g., bus voltage, branch power flows, loads and generations)	SPORF	The study presents an innovative data-driven real-time pre-fault DSA scheme using sparse projection oblique randomer forests (SPORF), encompassing offline training with feature selection, periodic model updates for adaptability, and real-time assessment based on PMU measurements	23-bus system and a practical 1648-bus system - 98.78% & 97.95%	2022
[77]	Dynamic	Active/reactive power (generation or load), power flows, voltages, and phase angles buses	CBDA C	The work proposes a novel conditional Bayesian deep auto-encoder (CBDAC) based DSA framework, consisting of off-line training, model updating, and online assessment stages	IEEE-68 Bus System - 95.8%	2021
[63]	Dynamic	Voltage magnitudes of the buses	DT	The methodology presented in this work combines physical system changes and data using a causality-based feature selection approach to create an efficient training dataset for enhancing data-driven DSA workflows in the face of network topology changes	IEEE 68-bus system - 91%	2022
[69]	Static	Phase angle and Magnitude of the bus voltage	MF	The methodology encompasses a continuous learning framework for online security assessment, integrating a weather-dependent security index, Mondrian partition-based online training with the Mondrian forest (MF) algorithm, copula-based correlation preservation, and a modified composite performance index	New England 39-bus and 118-bus standard test systems – 98.25%	2022
[78]	Static	the anticipated power supply and the operational status of components under an N − k contingency scenario	MLP	The methodology focuses on calculating the Expecting Load Shedding Cost (ELSC) for various contingency scenarios, with input data derived from stochastic simulations of N − k conditions. Optimal power flow algorithms are then employed to minimize ELSC for each scenario	IEEE-39 bus test system and PEGASE-1354 system – 98%	2024
[79]	Static	The active power generation of generators, the active and reactive power consumption of loads, and active and reactive flows of tie-lines	HGSA-FIS	The approach suggests FIS tuned by hybrid genetic-simulated annealing (HGSA-FIS), a hybrid genetic-simulated annealing tuned data-driven FIS, for quick SSA during single component failure. Evaluated on RTS-GMLC, it outperforms ANFIS and ANN in various scenarios	RTS-GMLC test system - APE< 0.05	2023
[80]	Static	Voltage levels at the respective equipment terminals and the voltage at the slack bus	FCM-ANN	The methodology combines fuzzy c-means (FCM) clustering and ANN classification to improve the accuracy and speed of static security analysis, using bus voltage data from offline simulations. It outperforms using ANN alone, though further enhancements are needed	The New England 39-bus system – 68.6%	2021
[81]	Static	voltage angle and magnitude	DT	The methodology comprises two key phases. Initially, discriminative factors characterize power system operating conditions. Upon encountering novel conditions, a sampling process updates the dataset through a Copula-based model, generating random points and running simulations for security labels	IEEE 39-bus system – 96.06%	2023
[82]	Static	Magnitudes of voltage, active and reactive power injections at systems nodes	CNN	The paper introduces a CNN-driven method for swiftly analyzing power system security under N-1 contingencies. It employs an automatic hyperparameter optimization technique and utilizes permutation feature importance to identify crucial features, reducing the initial input dimensionality	the simulation model of a practical grid in Mexico – 98.88%	2022

where AI has found significant applications. Voltage stability is the power grid adeptness to maintain the voltage of buses within permissible ranges subsequent to a disturbance arising from a specified operating condition.

TABLE 3. A summary of references on security assessment.

ML and DL based approaches	Other methodologies and hybrid techniques
TL [68, 75]	HGSA-FIS [79]
CNNs [73, 74, 82, 83]	FCM-ANN [80]
RFs [60, 69]	FIS [62]
AEs [67, 77]	Pattern discovery based fuzzy classification [84]
Extreme learning machine (ELM) [85, 86]	Radial basis function neural network (RBFNN) - PSO [87]
DTs [63, 64, 81]	ANN-PSO [66]
GP[49]	Fuzzy rule-based classifier with DTs [88]
GANs[65, 72]	
ANNs[89, 90]	
SVM[91, 92]	

To mitigate the potential for instability, the conventional approach to stability evaluation, frequently adopted in offline mode, relies on time-domain simulations involving selected contingency sets defined for anticipated operative conditions. However, this methodology has the drawback of being a timeintensive analytical process. Unregulated market activities introduce irregular power transfers or conditions challenging to predict within the systems. In the context of renewable energies, offline methods prove inadequate in representing the real operative conditions of wind and solar plants, leading decreased in system reliability. Therefore, traditional stability assessment techniques, reliant on offline mode studies, prove inadequate and contribute to heightened operational costs [1]. Driven by the need for reliable and secure power system operations, Operators are shifting from offline to real-time methods for voltage stability assessment. This transition involves the development of new, more accurate methods for real-time evaluation. One of the most critical prerequisites in real-time stability assessment is the availability of fast measurements of the system conditions, a challenge solved by the emergence of PMUs.

Depending on the time frame, voltage instability crises can be classified into short-term and long-term categories: Short-term voltage stability (STVS) problems occur within a brief time frame (seconds) following fault clearance. Long-term voltage stability- in contrast to STVS, involves a more extended time frame for instability issues or collapse, typically ranging from 0.5 to 30 minutes on average [51].

Stability indices play a pivotal role in the assessment of the power systems' voltage stability. These indices, characterized by scalar magnitudes, are employed to monitor variations in system parameters. Additionally, they quantify the proximity of a specific operating point to the point of voltage collapse [94]. STVS indices are used to monitor and manage

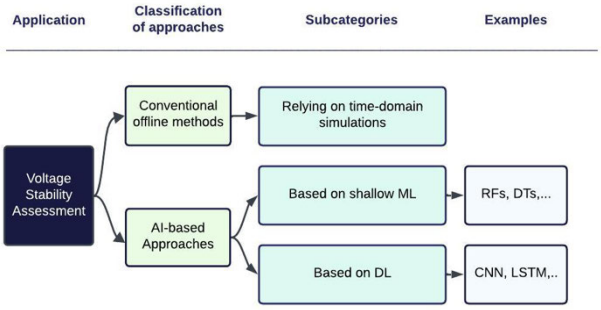


FIGURE 8. Classification of voltage stability assessment approaches explored in this paper.

the stability of a power system during dynamic events and transient disturbances, typically over short timeframes. They serve as early warning indicators, enabling operators to make real-time decisions to prevent voltage instability. An example of these indices is the instantaneous voltage index (VI) for each bus [95]. There is various voltage stability indices used in power system analysis, including the fast voltage stability index (FVSI), three diagonal element dependent index (l_{pi}, l_{qi}, l_i) and line stability factor (LQP) [96].

For LTVS, some of the indices assessing LTVS are voltage stability load index ($VSLBI_1$), voltage collapse proximity index (VCPI), voltage stability load bus index $VSLI_i$), and, simplified voltage stability index ($SVSI_i$) [97].

AI-based approaches have emerged as a solution to speed up the problem of stability assessment, as shown in Fig. 8. Next section covers voltage stability techniques using shallow ML, followed by DL methods.

1) VOLTAGE STABILITY ASSESSMENT BASED ON SHALLOW MACHINE LEARNING

In the independent study of STVS, shallow ML-based methods such as DTs, RF, and ELM are fast in assessment yet possess a restricted ability to extract high-dimensional nonlinear features. In [2], the authors employ ML for real-time STVS assessment using PMU data. It creates a time-series dataset, applies gradient-based data preprocessing, and combines RF and Light GBM (LGBM) classifiers. In [98], a method using DTs is introduced to classify numerous power system operating conditions based on voltage stability, with a new flexible criterion that identifies conditions near voltage instability regions while considering operational requirements. In [95], a novel methodology using data mining and ML is presented to assess STVS under significant disturbances, classifying power system stability states based on Lyapunov exponents and voltage indices, aiming to identify both fast voltage collapse and fault-induced delayed voltage recovery events in real-time. In [99], a self-adaptive data-analytics method is introduced for real-time STVS assessment, encompassing fault-induced delayed voltage recovery and voltage instability, optimizing accuracy, speed, and comprehensiveness through a multi-objective approach. In [100], a novel temporal-adaptive intelligent

system utilizing PMU and a diverse ensemble of randomized learning algorithms is introduced for optimized real-time assessment.

In the context of long-term voltage stability (LTVS), [101] uses a ML approach to predict LTVS by employing various voltage stability indices as inputs and conducts a comparative study to select the Random Forest Regression algorithm. The studies in [102] and [103] employ innovative hybrid approaches that synergize diverse methodologies. In [102], a hybrid approach combining the strengths of the dragon fly optimization algorithm (DFO-a metaheuristic optimization algorithm) and SVR is employed. DFO fine-tunes the parameters of the SVR model, enhancing its performance for the specific task of voltage stability assessment. The study in [103] also uses a hybrid of metaheuristic optimization and ML (ANFIS); ANFIS combines fuzzy logic and neural networks to create a model that can handle complex patterns and relationships in data. To explore more extensive details, [5] review ML techniques specifically applied to online voltage stability assessment. They aim to provide a comprehensive overview of the latest ML techniques used for assessing voltage stability in real-time scenarios within electric power systems. Further illustrations of the application of shallow ML techniques in the context of voltage stability assessment can be found in Table 4.

Shallow learning methods have several limitations. They struggle to fit complex functions, especially in highdimensional spaces, leading to reduced accuracy and performance. This is known as the ''curse of dimensionality.'' Additionally, their generalization ability is restricted, making them less effective in addressing intricate real-world problems. DL overcomes the limitations of shallow learning methods in several ways [10]:

- DL networks incorporate multiple hidden layers to describe data features with multi-stage transformations, resulting in superior feature expression, powerful function fitting, and enhanced generalization.
- DL models derive initial parameters from extensive data training, enabling independent feature extraction without manual intervention.
- DL methods maintain the integrity of information to a high degree and offer effective feature expression for complex data characteristics, making them well-suited for intricate data problems.

Hence, the utilization of DL in stability assessment has progressed rapidly. The next section studies Commonly used DL models in the context of voltage stability assessment.

2) VOLTAGE STABILITY ASSESSMENT BASED ON DEEP LEARNING

In light of the previously discussed advantages of DL, it is noteworthy that numerous academic works have incorporated DL techniques within the domain of voltage stability assessment. In another work [104], a DL model predicting STVS online by combining GCN for spatial features and LSTM for temporal features is presented, demonstrating its high performance through simulations in power systems. In [105], deep reinforcement learning (DRL) with a GCN is utilized to enhance power system voltage stability control by effectively capturing and adapting to topological changes and spatial correlations in nodal features, outperforming traditional fully-connected neural networks. In [106], a DL approach with an AE model for LTVS assessment is utilized, featuring LSTM networks and fully-connected layers, to assess real-time voltage stability using secure state data only, effectively evaluating stability through reconstruction residuals and achieving high accuracy in comparisons with traditional ML methods.

Traditional DL-based approaches for voltage stability assessment encounter challenges related to adapting to changes in the power system's topology, labeling samples effectively, and handling small datasets. Therefore, [107] utilizes deep TL to improve STVS assessment, employing PMU measurements, least square generative adversarial networks (LSGANs) for data augmentation, and a transformer model's self-attention mechanism to enhance model accuracy and adaptability to topological changes in power systems. Also, [108] employs TL by adapting a pre-trained model using only a few labeled samples to assess voltage instability under unseen network topology change conditions, enhancing the method's adaptability and real-time applicability in STVS assessment. In [109], a TL-based transient voltage stability assessment with small samples is proposed, effectively enhancing the evaluation performance of real-time power system transient voltage stability when sufficient historical data is not available.

Ultimately, the study in [10] provides an overview of the latest research on data-driven STVS assessment methods, emphasizing the use of both shallow and DL models, and discusses potential solutions to challenges in data-driven STVS assessment.

D. LOAD SHEDDING

Power outages have persistently posed a significant challenge to power systems worldwide, exerting a direct and consequential impact on the overall productivity within industrial sectors and various socio-economic domains [115]. Voltage instability, regarded as a principal contributor to power outages, stems primarily from power generation and transmission capacity inadequacies. In cases where the transmission infrastructure is insufficient to support the regular operation of the power system, load-shedding strategies are employed to mitigate excessive stress on the transmission network [116]. Although total elimination of blackouts due to unpredictable events and human errors may be unattainable, efforts are underway to reduce their frequency and duration through innovative methods and technologies. Consequently, researchers and utility enterprises have devoted substantial endeavors to prevent voltage collapse induced by natural occurrences and technical anomalies [117].

TABLE 4. AI-based voltage stability assessment methods in recent papers.

Ref	Type	Input data	Method	Summary of works' methodology	Test system and accuracy	year of study
[110]	ST	Normalized voltage amplitude, Active power, Reactive power	CWGAN-GP	The methodology involves labeling generated data samples using Semi-supervised Fuzzy C-Means (SFCM), and employing Conditional Wasserstein Generative Adversarial Network with Gradient Penalty (CWGAN-GP) to create a balanced training set. The Stability Assessment Transformer (StaaT) is then trained on this dataset and evaluated for performance against class imbalances and noise.	99.82 % on IEEE-39 bus system	2024
[111]	ST	Post-failure voltage time series data from three-phase fault simulations.	CNN-LSTM	The methodology involves generating post-failure voltage time series data through three-phase fault simulations in the IEEE Bus14 system. This data trains a CNN-LSTM model to forecast and categorize STVS incidents.	97.93% - The Kerala Grid at 220 kV	2024
[105]	ST	Bus voltages	GCN Based DRL	This letter presents a GCN-based method that combines topology-embedded raw feature extraction using GCN with sequential control decision-making using Double Deep Q-Network (DDQN) to mitigate voltage stability issues.	Tested on IEEE 39-bus system	2021
[112]	ST	Voltage trajectories of power system buses	MLP based DL model	The paper uses a multi-layer perceptron (MLP) DL-based approach to predict STVS without relying on post-fault data. It only utilizes direct PMU measurements at load buses, minimizing the need for additional measurements and computations while also demonstrating resistance to noisy data.	99.24% - tested on IEEE 39-bus system	2023
[100]	ST	Post-fault voltage trajectories	RVFL and ELM	The paper uses an integrated model of random vector functional link (RVFL) and ELM for STVS assessment and optimizes STVS accuracy and speed through a multi-objective programming framework.	99.36% - on New England 39-bus system and 99.21% - tested on Nordic test system	2020
[113]	ST	Voltage magnitude, active and reactive power injection	spatial–temporal GCN	This paper develops spatial–temporal GCN to incorporate spatial-temporal characteristics of short-term voltage instability. It uses Chebyshev graph convolution for topology, one-dimensional convolution for temporal features, and spatial-temporal information blocks, culminating in the assessment result through node and system layers.	98.7% - tested on a real-world Power Grid in China named Guangdong	2021
[107]	ST	normalized voltage, active and reactive power	Deep TL	For initial dataset creation the approach uses PMU data, for sample labeling uses temporal ensembling, for data augmentation employs lsgans, enabling effective DL on small datasets, and improving adaptability to topological changes by exploring fault connections.	98.47% - tested on IEEE 39-bus system	2023
[102]	LT	voltage magnitudes of the PMU buses	DFO and SVR	The study combines the DFO algorithm with SVR to optimize SVR parameters for real-time voltage stability assessment using PMU data, outperforming the alternative ANFIS model in predictive accuracy.	RMSE = 0.0273, IEEE 30-bus system and RMSE = 0.0565, Algerian 59-bus system	2018
[103]	LT	voltage magnitudes and phases from the weak buses	ANFIS with MFO	This study uses the ANFIS to assess voltage stability by considering voltage magnitudes and phases from weak buses. Weak bus identification is optimized with the moth-flame optimization (MFO) algorithm.	RMSE = 5.24* 10-4, tested on IEEE 30-bus & RMSE = 1.42*10–4, tested on IEEE 118-bus	2019
[106]	LT	Angles and Voltage magnitudes	LSTM-AE	This paper introduces an AE-based method using only secure data. The approach relies on the AE's ability to accurately reconstruct secure data and detect deviations, with LSTM networks and fully connected layers enhancing accuracy and a moving strategy biasing testing data towards secure features for robustness	precision:0.9792 - tested on IEEE 30-bus, 57-bus, 118-	2020
[114]	LT	Angle and nodal voltage magnitude of load buses	eGBDT	The paper proposes enhanced Gradient Boost Decision Tree (eGBDT) from high-dimensional PMU measurements. By applying GOSS and HEEFB, it achieves spatial dimension reduction and feature compression. The method also proves robust against missing data and noise compared to other approaches.	99.13% - tested on IEEE 118-bus system	2023
[1]	LT	voltage phasor synchronized measurements of certain buses	KELM	The paper proposes a novel methodology based on Kernel Extreme Learning Machine (KELM) to estimate stability status, foresee voltage instability due to reactive power limitations, and detect deficits in reactive power supply within system sub-areas.	tested on IEEE 39-bus system	2021

Both electricity demand and generation are influenced by the system's frequency and voltage. A decrease in frequency typically indicates a deficiency in real power, occurring when real power generation fails to meet demand. On the other hand, voltage collapse or voltage instability occurs due to inadequate reactive power supply from transmission lines and generators, leading to several major system failures globally [118]. Therefore, maintaining both voltage and frequency within acceptable limits is crucial for a stable power system. Under- frequency load shedding (UFLS) and under-voltage load shedding (UVLS) are deployed when frequency or voltage falls below set thresholds [119].

To mitigate frequency deviations induced by substantial loss of generation events, modern power systems employ a technique involving the controlled disconnection of a suitable portion of the load, commonly referred to as UFLS [120]. Indeed, UFLS demonstrates heightened effectiveness in electrical islanding, wherein the load reduction stabilizes the frequency within the isolated system [121], [122]. Several comprehensive literature reviews have been conducted within the realm of UFLS [123], [124], [125]. In [124], the authors review the UFLS methods applied in micro-grids (MGs), which still face power shortage challenges. They cove various aspects, including the MG system, frequency regulation and response analysis in MGs, and the impact of renewable energy resources on LS. The study in [125] reviews UFLS literature and algorithms using a systematic clustering approach and identifies research trends gaps in UFLS algorithms. Also, [123] reviews the literature on UFLS schemes is and introduces a new category focused on UFLS schemes involving smart loads while also discussing the potential of smart loads for frequency regulation and preventing UFLS.

UVLS strategies safeguard power grids against the risk of voltage collapse during contingencies [126]. Under specific circumstances, corrective measures such as shunt compensation and secondary voltage controls may prove inadequate to prevent voltage collapse. In such scenarios, the UVLS scheme may be implemented as a final measure to mitigate voltage instability [127].

Several in-depth literature reviews have also been undertaken in the UVLS field. In [128], a comprehensive review of UVLS is provided, explaining its crucial role in preventing voltage collapse in power systems, summarizing its principles, and discussing conventional and computational intelligence-based UVLS schemes. The study in [129] reviews existing UVLS tools and the application of hybrid computational intelligence techniques in addressing voltage instability in power systems, focusing on the strengths and limitations of different UVLS methods.

1) TECHNIQUES FOR LOAD SHEDDING AND THEIR VARIETIES

After 50 years of searching for the best or most optimal solution, a large and diverse set of UFLS and UVLS algorithms has been developed [130]. Present research regarding

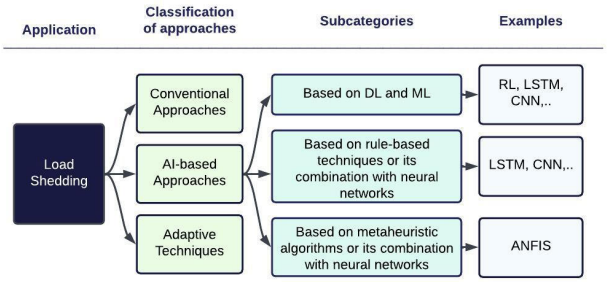


FIGURE 9. Classification of load shedding approaches explored in this paper.

LS techniques can be categorized into conventional, adaptive, and AI-based approaches, as depicted in Fig. 9. The location, time, and amount of LS represent three fundamental components within LS methodologies, and they exhibit a close correlation with and exert a direct influence on control performance [131].

Optimizing the amount of load is paramount when implementing LS procedures. Conventional LS methods involve a static approach, shedding a predetermined load amount in response to voltage or frequency deviations [120]. Nevertheless, this fixed approach may not always be optimal, potentially leading to either excessive or inadequate LS and, as a result, failing to establish an optimal LS scheme [132]. Adaptive LS techniques [133], [134], [135] dynamically shed varying loads in response to the system's disturbance level. Nevertheless, these methods may encounter suboptimal LS results from imprecise estimations of the system's diverse behaviors. The scope of this paper is predominantly centered on AI-based techniques, thus excluding conventional and adaptive LS methods from the primary focus of our research. AI-based techniques encompass a range of sophisticated methods, such as ANN, GA, fuzzy logic control, ANFIS, PSO, and numerous other cutting-edge approaches, all designed to leverage AI for various applications.

Numerous scholars have employed ANN in the context of LS within power systems [136], [137]. Nevertheless, it is important to note that ANNs tend to yield favorable outcomes primarily for well-defined and trained scenarios, displaying limitations in accurately predicting results for unknown or variable cases. Numerous scholars have employed fuzzy logic control in LS applications. In [138], a UFLS scheme utilizing fuzzy logic for dynamic adjustment of activation thresholds is introduced, aiming to enhance performance during substantial disturbances while minimizing power disconnections during minor and moderate disturbances in power systems. Hybridizing ANN and fuzzy logic, [139] presents a study on an ANFIS-based UFLS controller in the Tamil Nadu power system. The goal is to minimize LS and compare its performance with an ANN-based controller, finding that ANFIS can reduce LS and potentially increase revenue by 7.5 million INR per hour.

Researchers have furthermore applied a range of metaheuristic optimization techniques for conducting LS. In contrast to traditional approaches, these algorithms are esteemed for their precision, adaptability, and robustness in managing intricate nonlinear systems [129]; existing methods such as PSO and GA have shown different strengths and limitations. In [140], the authors propose a hybrid approach combining ANN with Back Propagation and GA to develop LSS in power systems to minimize LS while maintaining system stability. Also, [141] employs GA by using a multiple-deme parallel GA (MDPGA) to design of LSS to prevent voltage instability. The primary limitation of GA, which hinders their use in real-time applications, pertains to their slow responsiveness. Studies have revealed significant computation time associated with GA for LS calculations [142]. PSO has demonstrated its effectiveness as a robust and fast method for addressing nonlinear, multi-objective challenges. This technique has gained widespread acceptance in the field of power engineering. In [143], the authors address the necessity of maintaining a power generation-consumption balance in complex industrial environments during grid contingencies. It introduces an inventive LS algorithm that combines PSO with forecasted load information, offering an economical substitute to established UFLS procedures, which is validated through practicality assessments employing operational data from an industrial facility in Romania. Another LSS utilizing a metaheuristic technique called modified discrete evolutionary programming (MDEP) is introduced in [132] to determine the optimal load to be shed to achieve power balance in controlled islanding scenarios. It is compared to conventional evolutionary programming (EP) and exhaustive search techniques, showing that the MDEP technique provides optimal LS with the shortest computational time. Other metaheuristic techniques proposed for LSS in the power system application include firefly algorithm [144], ant colony optimization [145], and ant lion optimizer [146]. These LSSs determine the optimal load and the specific location to be shed in order to uphold the secure operating state of the power system.

2) REINFORECEMENT LEARNING FOR LOAD SHEDDING

Lately, DRL methods, which merge the state-perception capabilities of DL with the decision-making ability of RL, have gained recognition as a promising approach to address LS control challenges. DRL represents an AI technique more closely aligned with human cognitive processes and is often deemed as genuine AI. Fig. 10 [147] illustrates the foundational structure of DRL. DL acquires target observation data from the environment and offers the current environmental state information. Subsequently, RL links the current state to corresponding actions and assesses their values according to the expected return [148]. This iterative interaction process transforms decision-making into a step-bystep procedure. A RL framework comprises an agent, which serves as the decision-maker, operating within an environment. The decision-making process in RL can be delineated as follows.

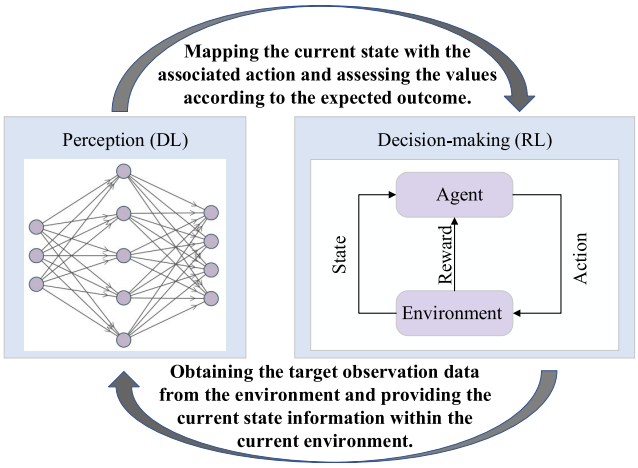


FIGURE 10. Foundational structure of DRL [147].

During each time step, the agent acquires an observation from the environment, subsequently rendering decisions by the observation following the current policy. The environment, in turn, responds to the agent's action, transitioning to a new state. and providing a reward signal for the evaluation of the action taken. The agent's objective revolves around maximizing rewards derived from interactions with the environment, which necessitates learning an optimal control strategy. RL can acquire optimal control behavior through persistent interaction with the environment and the gradient determined by the feedback reward signal [149]. DQN is utilized in [150] for Load-Shedding to tackle dynamic load-shedding challenges in extensive power grids. This approach employs a Convolutional LSTM (ConvLSTM) network to account for both spatial and temporal factors within the power system, ultimately enhancing voltage recovery performance. The DQN is a renowned DRL algorithm that excels in learning discrete action strategies but struggles with continuous action strategies.

Hence, [159] employs power system operation data to design a novel reward function based on voltage. It utilizes DRL with the Deep Deterministic Policy Gradient (DDPG) algorithm, a continuous training method, to formulate a LSS for controlling grid voltage during power system emergencies. In [160], a novel derivative-free DRL algorithm, PARS, is designed to address computational inefficiency and scalability challenges when applying DRL to power system voltage stability control with LS. The studies in [161] and [162] also propose a knowledge enhanced DRL method for intelligent LS. Table 5 compiles a comprehensive list of references encompassing AI-based LS approaches, providing a thorough repository of sources on this subject.

Finally, owing to the inability to store prior learned knowledge, DL models must be re-trained when facing tasks in a new scenario, leading to a long re-training process. Furthermore, the prediction accuracy is low due to the absence of sample data for new scenarios. TL addresses these challenges in [154] by harnessing the features learned to improve the performance of pre-trained DL models in new tasks.

TABLE 5. AI-based load shedding methods in recent papers.

Ref	Type	Input data	Method	Summary of works' methodology	year of study
[151]	UVLS	generator's active and reactive power, voltage parameters, the active and reactive loads, transmission line current	Risk-averse DL method	This article presents a risk-averse DL approach for real-time emergency LS which involves training a deep neural network to minimize the occurrence of load under-cutting events, which can lead to post-control instability and higher costs.	2022
[152]	UVLS	Load data (active and reactive power flow of buses, voltage at buses, the line reactance value, etc.)	hybrid algorithm based on GA and PSO	The paper uses a hybrid algorithm to optimize LS in response to contingency states, utilizing the FVSI to identify weak buses and significantly improving voltage profiles with reduced LS and faster convergence compared to GA alone.	2018
[150]	UVLS	Both competitive Q-network and Q-network input = current state	DQN	The work's methodology involves the development of a DQN for load-shedding that considers spatial and temporal information of a dynamic power grid using ConvLSTM networks and a novel reward function design, aiming to provide real-time LS decisions.	2022
[141]	UVLS	The load buses' active and reactive powers, AVR voltage output, transient direct axis and quadrature axis EMF of generators, and PQ bus voltages.	GA and neural networks	The work methodology involves using MDPGA to design LSS, triggered by a low voltage stability margin index, with computational speed improvements from a modular neural network and precise training through a simultaneous equilibrium tracing technique.	2021
[153]	UVLS	Bus voltages, active and reactive power Acquired by PMUs from the buses	deep feedback learning machine (DFLM)	The methodology employs a data-driven UVLS scheme, integrating a DFLM for precise prediction of voltage violations post-UVLS, with adaptive decision-making aided by stepwise constraint relaxation and incremental DFLM adaptation strategies	2021
[116]	UVLS	Line reactance and impedance, voltage and reactive power at the sending end, line loss	EP	The methodology involves the development of a computational intelligence-based technique, specifically EP, to ensure power transmission systems operate within acceptable voltage limits	2019
[154]	UFLS	Active power of each generator and load when the disturbance occurs (t0). Voltage parameters of each node at t0. Unbalanced power of each generator after the disturbance occurs (tf)	a TL and DL-based approach	The paper employs a two-stage DL method combining CNN and LSTM networks to forecast power grid frequency after disturbances, with an emphasis on spatial and temporal data, and it utilizes TL to adapt to varying data conditions for better real-time control and accuracy.	2021
[155]	UFLS	Spinning reserve, frequency decay rate, total power import, demand and in-plant generation	ANN-based technique with the regrouping PSO (RegPSO)	This article introduces an ANN optimized by the RegPSO algorithm for LS in an oil refinery power distribution system, aiming to improve the accuracy and speed of LS decisions compared to conventional schemes and Levenberg–Marquardt back-propagation ANN-based approaches.	2023
[156]	UFLS	The spinning reserve of each generator, the gains of the speed regulation, the amount of the active power imbalance, the inertia constants, and the load damping coefficient	stacked denoising auto encoder (SDAE)	The methodology involves using SDAE, trained through a two-stage strategy that includes pre-training and fine-tuning, with the incorporation of the dropout technique, used to predict post-fault frequency nadir and UFLS triggering stages. The model is fed with actual feature data as inputs, establishing a nonlinear mapping between inputs and outputs	2019
[157]	UFLS	Normalized fitness of the current best position in the t-th iteration for a specific particle. And normalized invariance for the best position of a particle in the t-th iteration for PSO.	PSO with Takagi-Sugeno fuzzy rules	This work involves optimizing shed load and frequency settings for an 81L relay in an offshore standalone power system with wind turbines. The objective is to minimize the total shed load and maximize the frequency nadir utilizing a TS-based PSO approach. This PSO method is fine-tuned with two sets of TS fuzzy rules for optimal parameter settings.	2022
[139]	UFLS	Frequency of the power grid, and the rate of change of frequency.	ANFIS	The methodology involves simulating and testing an ANFIS-based UFLS controller using back-propagation and least-square estimation algorithms in the MATLAB toolbox for the Tamil Nadu power system	2020
[158]	UFLS	Actor network input = current state Critic network input = current state + action output of actor network	DRL	The work's methodology involves formulating the emergency frequency control problem as a Markov Decision Process and addressing it using the distributional soft actor critic method, which estimates the distribution of the value function over returns for improved frequency control performance.	2022

E. OTHER APPLICATIONS

The subsequent section of this review paper broadens the scope by exploring additional applications within the domain of AI in power systems. This segment is concise yet enriched with the latest studies, offering a clear explanation of key conceptual frameworks that

define their integration with AI in current power system research.

1) FAULT DETECTION AND FAULT LOCATION

Despite the advancements in modern power systems, faults caused by various random factors remain frequent and inevitable. These faults significantly impact the performance of power grids, disrupting energy delivery and compromising system health, efficiency, and reliability. Consequently, the primary focus today is on the swift and accurate identification of these faults to facilitate their rapid elimination. This approach enables the maintenance of the faulty element and minimizes downtime by promptly restoring the system's primary functions. Moreover, it contributes to keeping component reliabilities closer to their initial levels [163]. Achieving this overarching goal necessitates the implementation of efficient protection and maintenance schemes.

Among the varied types of faults, electrical short circuit faults (SCFs) are noteworthy, as they may occur within generation, transmission [164], or distribution systems [165], [166]. Electrical SCFs exert detrimental effects on the operation of power grids, leading to the loss of their primary function [167]. In the occurrence of SCFs, various system signals, including voltages, currents, and rotor angles, undergo transient variations in amplitude, phase, and frequency. These variations are contingent upon the type and location of the fault. The temporal patterns of these signal alterations serve as a collection of covariate features that can be extracted to differentiate between SCFs and other disturbances in the system. In [168], transient variations in grid-connected wind turbine generators is tackled through proposing an intelligent relaying scheme that uses variational mode decomposition (VMD) and a DCNN to detect and categorize transmission line faults accurately. In [169], a MG fault detection and location module is proposed, utilizing one cycle of three-phase voltage and current measurements during faults as inputs developed with ML techniques, including Gaussian process regression for fault location prediction and SVM for fault detection. In [170], deep belief with buffalo optimization (DB-BO) algorithm is employed to identify and categorize faults in grid-connected PV systems, using power loss, total harmonic distortion (THD), and voltage deviations as input data. The studies in [171], [172], [173], and [174] provide additional examples showcasing the application of AI for fault detection and fault location in power grids.

The advent of wide area measurement systems (WAMs) facilitated by the emergence of PMUs has led to the availability of massive datasets. This abundance of data has contributed to the increased prevalence of data-driven approaches within power systems. In [175], advanced DL models (RNN and LSTM) is utilized to detect, classify, and precisely locate Frequency Disturbance Events in power grids utilizing limited PMU data. Modern power grids are transitioning to a digital and data-rich paradigm, with PMUs enabling the measurement of numerous variables, including current, voltage, and frequency, at various terminals. These measured variables serve as inputs for intelligent fault detection and classification techniques.

The study in [163] also utilizes three Deep RNN (DRNN) models based on DL for identifying fault regions, classifying fault types, and predicting fault locations in large-scale multi-machine power systems. Using PMU data, the models employ Sequential DL through LSTM to achieve accurate fault detection, classification, and location. Recent research endeavors leveraging the introduction of PMUs focus on addressing challenges related to fault detection, classification, and location in transmission lines by utilizing measurements from different points within the power grid.

2) POWER QUALITY DISTURBANCE

The modern power grid has become notably more intricate and diverse due to the integration of numerous new energy sources and the increasing variety of power load types. The introduction of new energy power generation brings about random fluctuations and intermittence. The uncertain and intermittent output from grid-connected new energy power generation systems can significantly jeopardize the stability of the power system [176], [177]. In an ideal power system, voltage and current waveforms exhibit sinusoidal characteristics at a constant 50 Hz or 60 Hz frequency. Any deviations from this ideal waveform are categorized as PQDs [178]. Various PQD events are recognized, such as voltage sag, swell, harmonics, spikes, flickers, transient interruptions, and others. These disturbances significantly affect the security and stability of smart power grids. Moreover, they can potentially result in malfunctioning of electronic controllers, programmable logic controllers, and smart relays [179].

Ensuring the adherence of grid power supply to established standards necessitates the immediate development of a proficient and automated technique for recognizing and addressing PQDs. This recognition not only identifies PQDs but also allows for implementing effective strategies to enhance power system stability. It involves the extraction of parameters by applying a specific signal processing technique, followed by subsequent classification [180]. The extraction of PQD features relies predominantly on experiential and statistical methods [181]. Some modern signal processing techniques include S-transform [182], wavelet transform [183] and Fourier transform [184]. Due to the inappropriate dimensions of certain features for classifiers or the time-intensive nature of classification, researchers often employ feature selection methods to address these issues in subsequent stages. A reduced-dimensional representation of the problem is obtained following the feature selection procedure [185]. Binary and multiclass classifiers are employed with those, as mentioned earlier, manually crafted features and feature selection techniques. The study in [186] proposes an approach for effectively identifying and classifying PQDs in distribution networks using an improved adaptive S-transform for feature extraction and RF for classification.

Also, [187] presents a novel approach combining Dual Tree Complex Wavelet Transform and SVM for precisely identifying and classifying PQDs.

DL, has arisen as a method, adept at autonomously acquiring optimal features from raw signals to enhance the process of classification. It addresses the challenge of PQDs caused by widespread nonlinear loads [188]. In [189], a 2D DenseNet is designed to learn relevant features from the raw signal data directly, eliminating the necessity for conventional feature extraction methods. Based on [190], 2D-CNNs demonstrate superior performance compared to 1D CNNs for feature extraction and enhancing model accuracy. The study conducted ablation experiments and found that the inclusion of 2D-CNNs significantly improved the average training accuracy, indicating their effectiveness in capturing more representative features. In [182], the authors propose an approach that incorporates an optimized S-transform for feature extraction and employ a CNN for automated feature selection and classification, demonstrating robustness in handling 37 PQDs detection and classification tasks, including single and multiple disturbances. The study in [191] also employs automatic feature extraction within a LSTM network, to detect and identify power quality issues.

DL methods have demonstrated auspicious outcomes, particularly in computer vision. Their performance has been comparable to, and in some cases, exceeded that of traditional ML methods reliant on manually crafted features designed by human experts [192]. The study in [193] aligns with the trend of DL's success in image analysis, showcasing its potential in PQ applications by employing TL for classifying PQDs, utilizing wavelet scalograms, and pre-trained CNN models such as VGG-19, Inception-v3, and ResNet-101. In [194], the authors conduct a comparative study to identify the most efficient DL method to classify PQDs. The study involves several models, such as ResNet-18, CNN+LSTM, LSTM, RCNN, and DNN, with ResNet-18 proving the most successful. GANs [194], [195], DAEs [196], and deep belief networks (DBNs) [197] have also been used for classification of various PQD events in various papers. Lastly, [198] provides an overview techniques for power quality analysis, including heuristic optimization, signal processing, AI, ML, and hardware implementation, with a comparative analysis based on criteria such as timing, accuracy, data conditions, and types of power quality events, offering valuable insights for researchers and power engineers specializing in power quality.

3) STATE ESTIMATION

Integrating a substantial volume of grid-connected renewable energy, coupled with the intricate transmission network and load variability, has led to heightened complexity in power systems' operation modes and dynamic characteristics. This complexity has resulted in increased randomness and volatility in power grids. Consequently, Power system dispatching departments endeavor to prioritize the rapid estimation of power system states, which is crucial for acquiring accurate real-time states and topological parameter information. This acceleration enables reliable dynamic evaluation and control, ultimately ensuring power systems' secure and stable operation. Power grid state estimation is the primary approach for obtaining real-time states to pursue this overarching goal [199]. State estimation entails estimating the actual system state variables, including voltage magnitudes and phase angles, utilizing raw measurement data gathered from PMUs and supervisory control and data acquisition (SCADA) [200].

a broad sense, State estimation is classified into two categories: dynamic state estimation and static state estimation. Since the real-time state awareness of the power system is a fundamental requirement for an effective energy management system (EMS), EMS acquires power grid data via SCADA, utilizing it to compute static state and power flow through power state estimation of power grid. Nevertheless, SCADA-based analysis and calculations primarily focus on quasi-steady state processes, and traditional state estimation falls short in delivering dependable, dynamic system states and quantities [201]. Consequently, it is essential to understand the roles of dynamic state estimation in power systems from different aspects. Dynamic state estimation plays a critical role in addressing this gap by offering a more comprehensive understanding of the system's dynamic behavior, thereby enhancing the accuracy and reliability of real-time state awareness. The latest proposed dynamic state estimation algorithms in recent literature include those based on RNNs [202] and LSTM [203] networks to model dynamic systems owing to their capacity to capture temporal dynamic behavior. Literature have also employes other ML-based solutions such as Adaboost [204], ANNs [205], SVMs [206], CNNs [207], and other AI-based methods such as FCM clustering [208], or ANFIS [209].

4) FALSE DATA INJECTION ATTACK DETECTION AND LOCALIZATION

The traditional power system is undergoing a transformation into a sophisticated smart grid driven by advancements in information and communication technology. This evolution entails the seamless integration of cyber and physical systems (CPS), giving rise to a highly interwoven CPS. The inherent interdependency within CPS underscores the importance of proactively preventing security attacks on information systems. Neglecting timely prevention measures can result in substantial repercussions, adversely affecting the safety, reliability, and economic functionality of the associated physical system [210]. FDIA, one of the most dangerous CPS attacks, was first proposed in 2009 by Liu et al. [211]. The article exposes an unknown vulnerability in current bad measurement detection algorithms by introducing and meticulously analyzing a groundbreaking class of attacks, termed ''false data injection attacks'', against power system state estimation, demonstrating the attacker's ability to manipulate measurements and systematically alter state

estimation results, even in scenarios with constraints on specific meters or limited resources.

Utilizing advanced intelligent algorithms, a data-driven model emerges as a viable solution for real-time analysis of grid operation data characteristics, enabling the detection of FDIA. This approach proves effective in addressing the elusive nature of FDIA. In [212], an SVM–Gentle Adaboost (SVM–GAB) algorithm is employed to identify FDIA through the effective dimension reduction and categorization of the measured data. In [213], a detection method for FDIA in smart grids is employed using a cyber-physical genes-based approach, unsupervised pre-training encoders for extracting features related to cyber-physical attack genes, and a SAE network for constructing an FDIA detection model. The model parameters are optimized with a self-adaptive cuckoo search algorithm. In [214], a real-time FDIA identification scheme based on LSTM is employed for its ability to model complex, nonlinear relationships in time-series data effectively. It is well-suited for forecasting and detecting FDIA in smart grids. The LSTM structure is highlighted for its superior performance to other approaches, such as MLP, SVM, and ARIMA. The study in [215] underscores the utilization of VMD for spatial and spectral feature extraction, followed by training an LSTM-AE to learn temporal correlations in multi-dimensional feature vectors. Additionally, a Logistic Regression classifier is employed based on error deviation vectors to determine abnormality in the proposed detection mechanism. The study in [216] utilizes graph edge-conditioned convolutional networks (GECCN) to detect FDIA in power system state estimation. GECCN leverages topology information, node features, and edge features to capture correlations in sample data effectively. In [217], DRL is employed to detect highly stealthy FDIAs in the smart grid.

In addition to FDIA detection, certain literatures also investigate the localization of FDIA within smart grids, too. In [218], the authors introduce an innovative method for localizing FDIA using graph data modeling and DL. Also, [219] proposes a novel method, employing cubature Kalman filter, Gaussian process regression, and a DCNN for identifying and localizing FDIA in power grids. In [210] another instance of utilizing DL for FDIA detection and localization based on a self-attention DCNN is employed.

For a more thorough understanding of FDIA in power systems, [220] provides a comprehensive discussion on FDIA in power systems, delving into the goals, construction methods, and consequences from the attacker's perspective while also examining protection and detection countermeasures from the defender's viewpoint.

IV. INTEGRATING ARTIFICIAL INTELLIGENCE AND DIGITAL TWIN IN THE NEXT GENERATION OF POWER SYSTEMS

Power systems are transitioning from a centralized and passive structure to a distributed and active configuration, driven by the need to meet increasing demand and expectations for low-carbon emissions. Integrating more renewable energy resources, the diminishing role of conventional centralized power plants, the widespread adoption of energy storage systems, the electrification of prosumers, and the evolving electricity markets collectively foster a versatile and dynamic power system. This transformation demands increased intelligence and digitization, leveraging extensive data to ensure its proper, secure, and secure operation in the future, all while capitalizing on emerging financial opportunities [221]. However, owing to the extensive diversity and substantial volume of data generated within modern grid systems, there is a pressing need for sophisticated solutions capable of seamlessly integrating, analyzing, and extracting essential and significant information. AI-based solutions offer a promising avenue for tackling the challenges stemming from the complex nature of modern grid systems, primarily by data mining, which facilitates the forecasting and optimization of grid operations [222]. AI-based solutions confer several benefits, including rapid processing, intelligent decision-making, precise classification, and accurate predictive results. The efficacy of AI-based solutions, nonetheless, relies on the accessibility of datasets and computational resources, enabling the training and generalization of the model for a particular task [223]. When incorporating AI applications into real-world power systems, it is imperative to contemplate various data-related considerations meticulously. AI applications are moving from being developed and tested in controlled laboratory environments to being used in real-world power systems; consequently, there are several important issues related to data that need to be addressed. Ensuring the accuracy and robustness of AI algorithms in the realm of power systems relies on access to sufficient historical data. While power systems generate substantial amounts of data, the availability of such data that aligns with particular AI requirements is notably constrained. Although open-source and synthetic datasets offer potential substitutes, their inherent limitation lies in their capacity to exclusively represent the unique features of the originating system, making their adaptability to various applications or systems a challenging endeavor. Modern power systems offer diverse data sources encompassing electrical attributes and behaviors and a comprehensive array of data types across various domains [224]. The intricate task of harmonizing and amalgamating these data types into a uniform format to accommodate a singular application is compounded by variations in data formats and protocols, even when dealing with data of the same category, necessitating preprocessing measures such as data cleaning and filtering. Finally, Data in power systems are gathered from various sources, including PMUs and SCADA systems, as well as a range of additional sensors; nevertheless, there exists variability in sampling rates, and some may not meet the requirements of specific AI applications. Moreover, environmental noise and uncertainties impact data quality, leading to synthetic and offline simulation data for AI training. However, these data may not fully replicate real-world conditions due to inherent

simplifications and assumptions in power grid models. The abovementioned issues pose critical obstacles to the broader adoption of AI in power systems. To address data-related concerns and promote the development of AI applications, it is essential to employ technologies that can effectively bridge the digital and physical worlds. Progress in computational technologies also facilitates the construction of comprehensive digital models of physical systems, commonly called digital twin [225]. The integration of AI algorithms and solutions can further enrich these digital replicas. The potential for combining AI and digital twin in power systems presents a significant opportunity to address evolving challenges and advance future enhancements.

A. CHALLENGES IN PRACTICAL IMPLEMENTATION OF ARTIFICIAL INTELLIGENCE IN POWER SYSTEMS

Leveraging AI applications holds the promise of substantially enhancing problem-solving capabilities through implementing more intelligent solutions.

Numerous AI-based solutions for power grids were suggested in previous sections, each offering a diverse array of distinctive features. These features are inherently aligned with the advancement and deployment of intelligent applications tailored explicitly for addressing issues within power systems. A selection of these features is summarized in Table 6 [223].

Despite the proliferation of AI applications in scholarly literature, the practical implementation of such applications in power systems remains constrained. In the following, we aim to introduce the challenges associated with the practical implementation of AI applications in power systems.

Firstly, the internal structure and logic of AI algorithms present challenges regarding explicability and comprehensibility, as users can only observe the inputs and outputs without insight into the intervening processes [226]. AI applications' constrained interpretability is significant when deployed in power systems and associated assets. This limitation could impact these systems' secure and secure operation, imposing constraints on human involvement in problem-solving tasks and elevating the risk of potential failures.

Secondly, substantial computational and storage resources are imperative for AI-based solutions in power systems, involving tasks such as managing large datasets and ensuring data availability at all times [227]. Traditional ML methods encounter difficulties managing the substantial computational demands, necessitating significant resources for training models and handling extensive datasets.

Thirdly, in light of the inherent intricacies within power systems and their components, creating a reusable test bed for the development of AI algorithms is challenging. Consequently, the optimization of algorithms is often tailored to address specific issues and meet the unique requirements of particular use cases. For instance, with significant changes in topologies driven by maintenance and control actions [63] or variations in operation conditions, such as significant load level fluctuation [228], the performance and accuracy of data-driven approaches diminish significantly.

Finally, disparities between simulated models and real-world systems persist as a result of the simplifications and underlying assumptions inherent in the latter, compounded by the uncertainty and complexity of real-world systems. Given that numerous AI-based solutions are formulated based on simulation datasets [229], AI models may face challenges when encountering unforeseen faults or significant changes in the power system [68]. The limited coverage of specific faults in simulations results in diminished performance and accuracy when faced with unforeseen or novel fault scenarios in real-world applications. Additionally, Traditional methods rely on complete data inputs, leading to inefficacy when faced with missing data [65].

In the previous sections, we discussed the scope of AI within power systems and its diverse potential applications. We also highlighted that data plays a crucial role as both a key supporter and a significant obstacle to applying AI solutions. TL emerges as a pivotal solution for mitigating some of the challenges. Its capability to harness knowledge acquired from one task or domain and apply it to another significantly improves model training efficiency, particularly in scenarios with limited data availability or domain shifts. Addressing the remaining challenges, integrating AI models with TL and digital twin technology proves to be a synergistic approach, proficiently tackling complexities inherent in real-world systems. This integrated methodology further elevates the adaptability, scalability, and precision of AI models fused with digital twin technology, transforming them into potent tools for comprehending and optimizing intricate physical systems. Digital twin, digital representations of real power systems, offer a promising solution to address numerous data-related concerns and barriers, making it easier to incorporate AI algorithms effectively.

A common assumption in ML-based solutions is that the training and testing set share similar data distributions. Consequently, many scholars managed to generate extensive training sets that encompass as many scenarios as possible. However, this assumption fails when applied to real-world, time-varying power systems, as the training set cannot encompass all potential scenarios. In time-domain simulations, the number of simulation cases increases exponentially with the number of random parameters, making it impractical to account for all variables in power systems [230]. Domain adaptation, a feature-based TL technique, addresses this issue by allowing the training and testing data distributions to differ [231]. It offers a compelling solution by mitigating the effects of domain shift without necessitating extensive retraining. These methods leverage the concept of aligning the feature distributions between the source and target domain data. However, significant disparities between these domains may still require alternative solutions.

TABLE 6. AI features.

Functionality	Definition	Examples of usage and reference
Pattern recognition	Automated identification of patterns and relationships within data and making predictions or classifications based on that knowledge.	• Fault diagnosis in transformers and motors [238] • Estimating the remaining useful life of batteries [239] • Predicting renewable energy consumption [240]
Decision making	Extracting useful features from collected information to support humans in making more accurate decisions or replace humans in taking actions to increase operational efficiency.	• solving decision-making problems in smart grids [241] • forecasting of renewable power outputs [242]
Parameter tuning	Automated parameter optimization based on the latest dynamic changes and status of the system	• using trained DL models of old PV plants in recently installed PV plants in the same neighborhood By leveraging TL [243] • Minimizing energy expenses for commercial buildings by accounting for variable electricity prices and demand over time [244]
Feature extraction	reducing raw data dimensionality without sacrificing essential or relevant information	• Extracting features of adjacent sliding-time windows for fault localization strategy in multilevel converters [245] • Using feature extraction for identification of spatial patterns and point information from satellite images to forecast solar radiation [246]
Data fusion	Combining various data sources to generate precise and consistent information, resulting in a more comprehensive description of the system.	• Integrating cloud images and radiation data to forecast solar energy, taking into account both spatial and temporal details [247]

In such case where the domain disparity is significant, parameter-based TL remains a crucial tool, eliminating the need to train models from scratch. Parameter-based TL involves leveraging a pretrained model, often trained on a large and diverse dataset (source domain), to solve a new but related problem (target domain). The process is typically achieved by freezing some of the initial layers of the pretrained model and fine-tuning the final layers using the data specific to the new task [232]. This approach offers several practical advantages. Firstly, starting with a pre-trained model, the initial weights are already tuned to recognize general patterns, which means the model is much closer to the optimal solution than if it started from random weights. This significantly reduces the number of epochs needed to train the model on the new task, leading to faster convergence. Secondly, using a pre-trained model means that many of the early layers can be frozen, meaning their weights are not updated during training. This reduces the number of parameters that need to be trained, lowering the computational resources required. Additionally, Parameter-based TL allows models to achieve good performance with a smaller dataset. Since the pre-trained model has already learned a robust set of features from a large dataset, it can apply this knowledge to the new task, requiring less data to fine-tune the weights and biases. In conclusion, Parameter-based TL streamlines the optimization of weights and biases by leveraging pre-trained models, which accelerates training, improves performance, reduces data needs, and enhances computational efficiency. This makes it a valuable technique in scenarios where time and resources are limited. For detailed information on how pre-trained networks are applied to different tasks and fine-tuned through a systematic process, refer to [233] and [234].

TL, while a powerful approach in ML, is not without its limitations. One significant constraint is the domain mismatch risk between the source and target domains. In cases where significant domain divergence exists between the source and target domains, retraining a model from scratch is warranted. In computer vision learning tasks, discrepancies often arise when there are fundamental differences in the data characteristics between the source and target domains, including variations in channel numbers, intensity levels, scales, and orientations [235].

The second limitation of TL pertains to its significant reliance on extensive pre-training datasets, crucial for enabling TL models to capture comprehensive features and achieve effective generalization [236]. Even with the abundance of large-scale datasets developed over time, their suitability across different domains varies. Certain

professional domains encounter persistent data scarcity issues that cannot be adequately resolved using widely available datasets like ImageNet, potentially introducing domain mismatch concerns. Additionally, utilizing smaller pre-training datasets may increase the risk of overfitting, reduce model reliability during target model training, and inefficiently utilize resources. Therefore, meticulous selection and augmentation of pre-training datasets are crucial to address these challenges and optimize the efficacy of TL methodologies. Poor selections and expansions may introduce undesired elements, leading to adverse transfer effects that compromise the overall effectiveness of the transfer process [235].

Besides the challenges, there are additional challenges related to integration, scalability, and interoperability with legacy systems in deploying AI within power systems. ML models have demonstrated significant potential, but many face limitations in understanding their uncertainty, known as confidence awareness. This concept involves the model's ability to quantify its confidence in outputs based on available data. This understanding is critical across various applications, enabling informed decision-making by acknowledging and appropriately managing uncertainty levels [77]. Scalability is another significant challenge of integrating AI into power systems, which refers to the ability of an AI model, developed and tested on a small-scale system, to effectively function when applied to larger, real-world power systems. Therefore, researchers test AI models on larger systems to ensure scalability, robustness, and real-world applicability. This practice demonstrates that AI models can manage the intricacies of real-world power grids, enhancing their reliability and suitability for practical deployment. Although testing on larger systems helps address some scalability issues, the challenge remains due to the inherent complexities and diverse scenarios of real-world power grids. Based on [237], the existing centralized control systems have limited flexibility and scalability, making it difficult to properly integrate and manage the growing variety and quantity of utility-owned and non-utility-owned digital devices and systems. Lastly, Interoperability, which refers to the ability of two or more devices, networks, systems, or applications to exchange and use information for correct cooperation, is another challenge in integrating AI into existing power infrastructures. In smart grids, interoperability must be well specified in use cases to develop systems that are interoperable by design, ensuring seamless information exchange and performance of required functions [248].

The following sections offer a concise overview of TL and digital twin technologies, detailing their fundamental components and technical workings. Emphasis is placed on their potential role as key enablers for integrating AI-based solutions into power systems and their associated assets.

B. CATEGORIZATION OF TRANSFER LEARNING

Offline training of DL models generally necessitates a substantial amount of labeled data and is often a time-intensive process. Furthermore, ensuring that the trained model performs optimally across all operating conditions of power systems is challenging. To significantly reduce computational costs and improve model adaptability, TL is extensively utilized in power system applications [228]. TL methods can generally be divided into three categories: parameter-based, feature-based, and instance-based methods [249]. Each of these categories will be discussed in detail in the following.

1) INSTANCE-BASED TRANSFER LEARNING

The instance-based TL method involves reweighting labeled data from the source domain to be applied in the target domain, presenting an approach that is straightforward, efficient, and theoretically comprehensible. The fundamental concept of these methods is to reweight the labeled training data to address the discrepancies between source and target distributions. This reweighting is implemented by multiplying the individual loss of each training instance by a positive weight during the training process [250]. The general framework of the instance-based TL is shown in Fig. 11(a).

2) PARAMETER-BASED TRANSFER LEARNING

The parameter-based TL method involves transferring knowledge at the model or parameter level. Implemented practically, this approach frequently involves freezing certain neural layers in the source domain model and retraining the final one or two layers using data from the target domain system [230]. Typically, these frozen layers are utilized to extract features from the input data. The extracted features are then mapped to the target state using various weights and biases in the fully connected layers. The parameter-based TL technique described in this paper is depicted in Fig. 11(b).

3) FEATURE-BASED TRANSFER LEARNING

Feature-based methods rely on identifying common features that exhibit similar behavior concerning the task across both the source and target domains. A new feature representation, commonly known as the encoded feature space, is created using a projecting application that aims to align the source and target distributions. The learning task is then performed within this encoded feature space [252]. The feature-based TL framework is depicted in Fig. 11(c).

C. CONVENTIONAL NOTIONS OF DIGITAL TWIN

The term "digital twin" was initially introduced in 2002 [253]. The concept has found application across diverse industries, with distinct interpretations. Its utilization within manufacturing, aviation, healthcare, and the energy sector has given rise to varying conceptual delineations of a digital twin and its potential utility [254]. Notwithstanding the variations in the particulars of these interpretations, a prevailing consensus concerning the fundamental constituents of the digital twin concept, comprises three core elements. Grieves [255] originally introduced digital twin with three key components for the first time: the digital or virtual representation, the physical product in the real world, and the

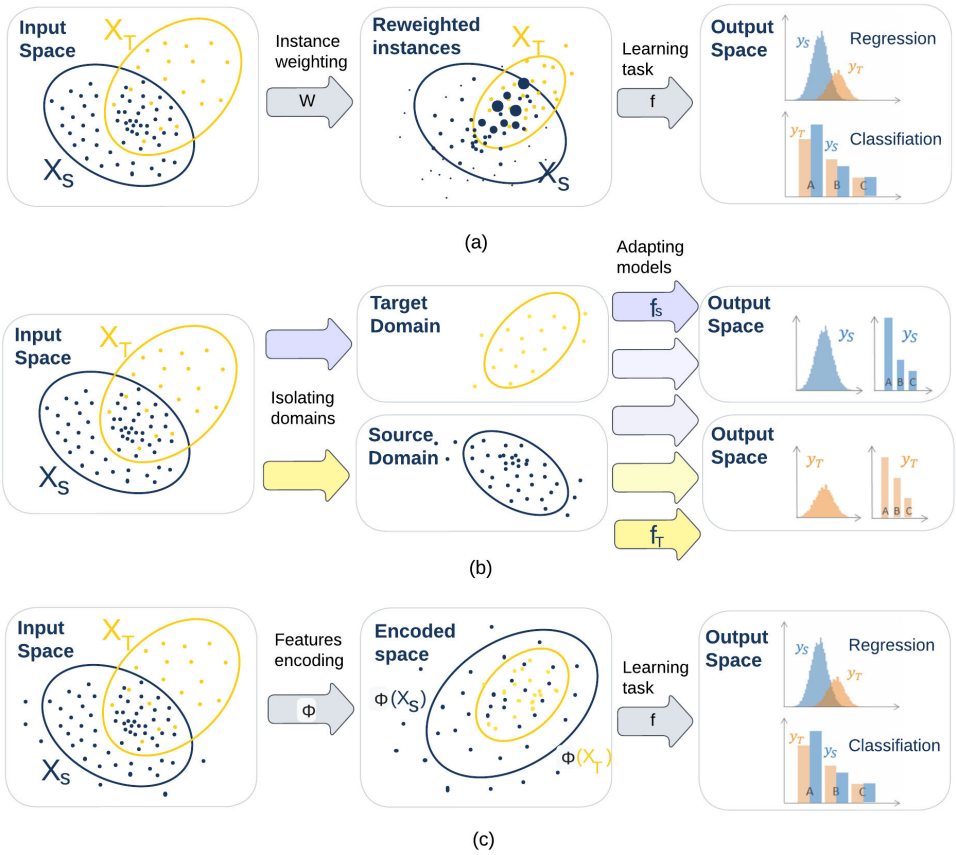


FIGURE 11. Transfer learning frameworks: (a) instance based transfer learning, (b) parameter-based transfer learning, (c) feature-based transfer learning [251].

connection bridging the two—nonetheless, some authors, like Tao et al., have expanded this concept to encompass five components in [256], which now include data and service as other elements of the Digital Twin. Tao et al. [257] further determine VV&A (verification, validation, and accreditation) as components of digital twin and emphasize that the seamless integration of both cyber and physical realms distinguishes digital twin." While the abovementioned definitions are conceptually valid, achieving a consensus on a precise Digital Twin definition necessitates delineating its fundamental prerequisites. As technology has evolved, with digital twin relying on ML, big data, and cybersecurity advancements, these requirements have undergone alterations. Furthermore, the domain-specific nature of digital twin underscores the need to establish components that can be universally applied across various domains while acknowledging that their extent of inclusion and measurement may vary depending on the specific domain [258].

D. A LIVING DIGITAL TWIN FRAMEWORK FOR POWER SYSTEMS AND ITS INTEGRATION WITH AI

A digital twin begins with the physical asset, the actual system or component in the real world being digitally twinned. The overall structure of a digital twin, including its key functional layers, is illustrated in Fig. 12.

A physical asset can be a single component, such as an individual generator (e.g., a wind turbine), an entire generation system (e.g., a wind farm), or even the complete electrical grid. The physical asset continuously gathers real-time operational data, including voltage, current, temperature, and environmental parameters, even as it experiences environmental stresses like load fluctuations, temperature extremes, and mechanical strain, all precisely measured by sensors and SCADA systems. Specialized control modules (CMs), typically integrated within local hardware or embedded in existing SCADA, FPGA, or DSP systems [259], receive commands from the digital twin to adjust the asset's operation. Thus, the physical asset provides ongoing precise measurements that directly influence the digital twin's accuracy and quality over the entire asset lifecycle.

Next, real-world data is transmitted through the connection layer. This layer is the digital twin's critical communication backbone and enables two-way communication between all the components. Using standardized and secure protocols (e.g., IEC 61850, Modbus, and IEEE C37.118) ensures a synchronized flow of real and virtual data measurements,

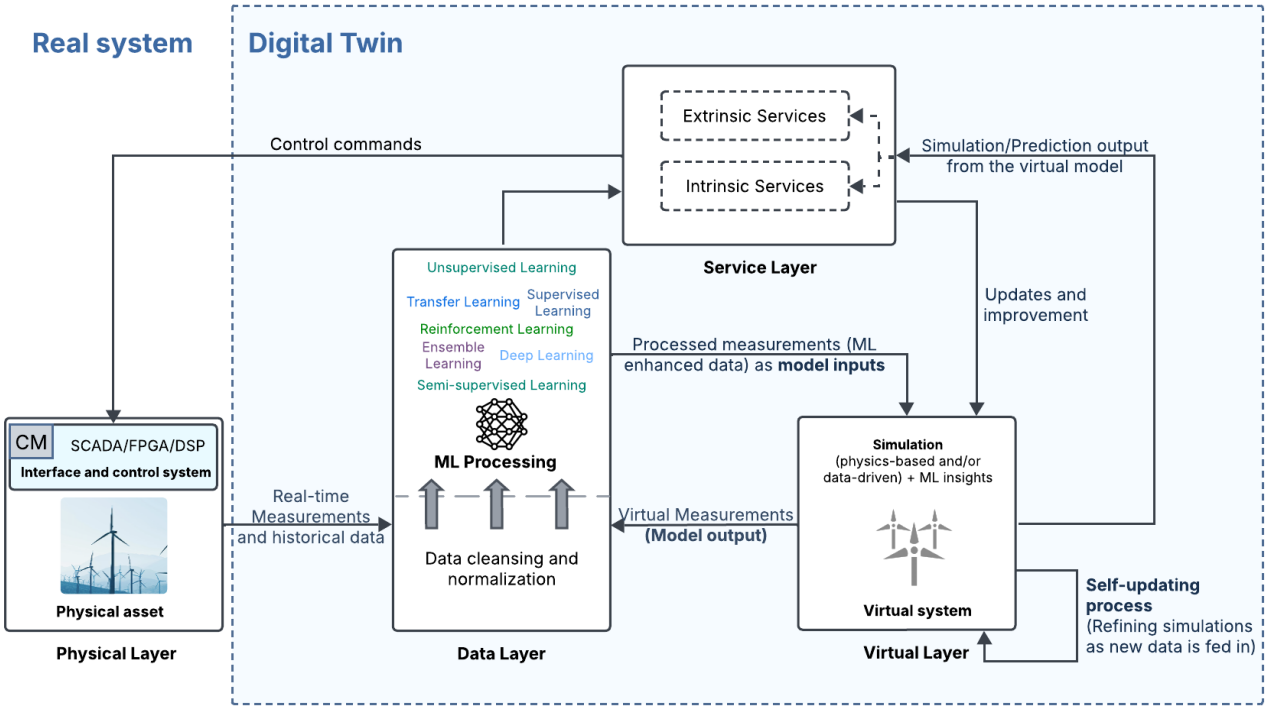


FIGURE 12. Integration of machine learning and digital twin.

control commands, and data management instructions. Real-time synchronization between physical and virtual assets without this communication infrastructure is not possible.

After transmission, measurements are received by the data layer. The data layer is the central repository responsible for storing and managing real and virtual measurements. This layer processes substantial streams of real-time sensor data and historical records, including electrical loads, temperature variations, and component statuses, through cleaning, aggregation, structuring, and archival procedures. The data analytics frameworks and ML algorithms are also used in this layer [260] to identify meaningful patterns within these datasets since they can facilitate predictive analytics and support informed decision-making. Consequently, processed data become immediately accessible to higher-level analytical services or virtual modeling components.

Using structured data, the virtual model replicates the physical asset by integrating real-time sensor measurements, physics-based modeling, and ML methods to reflect current operating conditions accurately. Thus, it is capable of predictive analytics and comprehensive scenario evaluations as the virtual model operates within a computational environment free from physical constraints [261], and engineers can virtually analyze scenarios such as load variations, generation changes, or component failures without incurring expenses or physical risk. The virtual model continuously updates its parameters in response to changes occurring in the physical asset to maintain accurate alignment with real-world conditions.

Finally, the service layer converts virtual model outputs andstored data into actionable intelligence and strategic decisions. virtual measurements, even though they originate from the virtual model, also undergo further processing for consistency and integration with other data inputs. The service layer comprises intrinsic services, the core functionalities that maintain real-time synchronization and accuracy of the digital twin through continuous calibration, data coordination, and clock synchronization. Intrinsic services maintain essential functionalities like data coordination, real-time model calibration, and clock synchronization. Reporting modules within intrinsic services interface externally for evaluation of performance. Extrinsic services provide specialized, value-added capabilities, such as predictive maintenance, condition monitoring, and advanced operational controls (e.g., economic dispatch, load optimization). Commands generated by the service layer are then communicated back through the connection layer, triggering adjustments in the physical asset's operation and completing the digital twin feedback loop.

Through this continual feedback loop (physical asset conditions feeding data into the digital twin, refined virtual models generating predictive insights, and the service layer executing optimal actions back into the physical world), the digital twin emerges as a dynamic, adaptive, and continuous learning framework. This iterative process ensures precise

TABLE 7. Key dimensions across each layer of a digital twin.

Dimension	Physical Asset	Virtual Model	Connection Layer	Data Layer	Service Layer
Role/Definition	Real-world entity (e.g., component, plant, or entire grid) + control interfaces (SCADA,FPGA, etc.).	Software or mathematical mirror of the physical system, using simulations/analytics.	A secure communication bridge for real time data and command exchange between all digital twin components.	Stores, manages, and organizes real & virtual measurements (historical/current)	Decision-making and advanced functionalities, split into **intrinsic** (core DT tasks) and **extrinsic** (added value)
Key Functions	-Generates live measurements. -Receives commands from DT services.	-generates simulated outputs (virtual measurements) and forecasts. -Receives virtual actions.	-Real time bidirectional data and command exchange. -gathering sensor data and converting various industrial protocols.	-Real time data acquisition, storage and analytics. -Massive data management with ML insights	-Intrinsic Services: Coordination, Synchronization & Calibration -Extrinsic Services: Monitoring, Maintenance & Control
Key Challenges	-Sensor reliability and accuracy. -Environmental stresses altering measurements.	-Maintaining high-fidelity in dynamic environments. -Synchronizing with the physical asset's state.	-Ensuring low-latency, secure data flow. -managing a high-volume of real-time data from diverse endpoints.	-handling the large volumes and variety of data from physical and virtual assets. - ensuring data quality and security.	-Maintaining robust core services while flexibly integrating optional features -Ensuring robust coordination across the DT
Key Components /Technologies	SCADA, IoT sensors, PMUs, FPGA, etc.	Simulators, Analytics and ML algorithms (to predict future states or simulate what-if scenarios).	industrial/IoT protocols (IEEE C37.118, IEC61850, etc.).	SQL/NoSQL Databases, data warehouses, data lakes, big data analytics	Advanced controllers, Data coordination mechanisms, Reporting modules, etc.

synchronization and ultimately enhances reliability, operational efficiency, and strategic management of power systems across their entire lifecycle.

E. PRACTICAL APPLICATIONS AND CHALLENGES OF AI-ENHANCED DIGITAL TWINS IN POWER SYSTEMS

The integration of AI with digital twin holds significant promise for revolutionizing applications and services within power systems. Combining the functionalities of digital twin and AI unlocks a spectrum of applications and services applicable to power grids and assets. This integration allows for the incorporation of multiple AI-based functionalities into digital twin, facilitating advanced data analytics and decision-making processes [223]. Fig. 12 illustrates a basic representation of integrating ML with digital twin, emphasizing the primary learning methods, including supervised, semi-supervised, and unsupervised learning, TL, DL, etc. The integration of ML methods within digital twin technology entails the implementation of diverse ML algorithms and techniques to enhance the functionality and efficiency of digital twins.

Reference [262] addresses the need for a comprehensive digital twin architecture for power plants, specifically focusing on integrating DL to enhance system predictability and anomaly detection in operational power plant data. While it recognizes the need for data-driven algorithms, it emphasizes the inadequacy of a standalone data-driven approach, suggesting a combination with a low-order physics-based model for better results. In [263], a digital twin framework that integrates AI-based security assessment methods is introduced to monitor and manage the complex challenges in modern power systems driven by increased renewable energy integration and evolving load types. In [264], CNNs have been incorporated into the Automatic Network Guardian for Electrical systems (ANGEL) digital twin environment for fault detection in power grids. The implemented framework demonstrates noteworthy precision in fault detection, and the system is designed in a way that allows it to be used for larger power systems without losing its effectiveness.

In [265], DL with a digital twin framework is combined to address challenges in modeling power grids' uncertain and dynamic characteristics, especially with the integration of renewable energy sources. The proposed model-data hybrid-driven concept within the digital twin framework improves the technique's effectiveness. The study employs the time convolution neural network algorithm to capture high frequency switching states and dynamic changes in renewable energy unit operations precisely. Reference [266] proposes a wind turbine controller using a digital twin with adaptive control facilitated by deep deterministic policy gradient, a DL approach used in RL.

The study by [267] reviews digital twins' evolution, properties, and variants in the electric power sector and explains their early applications. It also highlights the transdisciplinary challenges that can be mitigated through co-simulation and

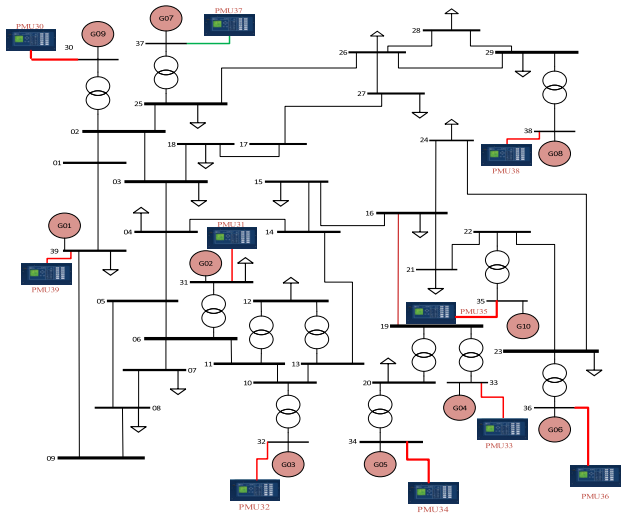


FIGURE 13. The IEEE-39 bus system.

TABLE 8. A summary of references on security assessment.

Fault number	Fault setting	Duration
F1	Fault bus 3, Trip 3-4	0.25
F2	Fault bus4, Trip 3-4	0.25
F3	Fault bus4, Trip 4-14	0.25
F4	Fault bus14, Trip 4-14	0.25
F5	Fault bus2, Trip 2-3	0.25
F6	Fault bus3, Trip 2-3	0.25
F7	Fault bus6, Trip 6-11	0.2
F8	Fault bus6, Trip 6-11	0.2

co-modeling, thereby guiding power system professionals in adopting these technologies.

Stakeholders within the power system sector are diligently striving to enhance the operational flexibility of power plants, reduce downtime, decrease operation and maintenance expenditures, and enhance overall efficiency, productivity, and profitability. Within this context, Digital Twin technology was introduced as a pivotal enabler to achieve these goals.

Despite significant research advancements in Digital Twin over the past decade, several challenges remain to be addressed for its successful implementation across diverse sectors. During the implementation phase, domain-specific Digital Twin initiatives encounter hurdles such as complexity, integration challenges, component volume, resource requirements, data scalability, performance metrics, interoperability issues, feasibility, and domain expertise.

Additionally, as discussed in section IV, despite the widespread adoption of AI applications in academic research, their practical deployment in power systems is hindered by challenges related to data, including availability, diversity of sources, quality, and temporal constraints. Overcoming these data-centric issues and advancing AI development necessitates technologies that effectively connect virtual and physical domains. This is where digital twins have emerged as pivotal innovations. However, the integration of digital twins with power systems also introduces cybersecurity challenges, particularly concerning data integrity, privacy, and availability. Therefore, ensuring robust cybersecurity measures alongside advancements in data integration technologies is paramount to harnessing the full potential of AI-enhanced digital twins in optimizing power system operations while safeguarding against cyber threats.

In summation, the practical implementation of AI in power systems poses several challenges, significantly impeding the pervasive application of AI methodologies within real-world power systems and impacting their secure operation. To surmount these challenges, the integration of digital twins with AI emerges as an imperative strategic recourse. The synergistic integration of AI, TL and digital twin not only mitigates extant challenges but also propels power systems toward heightened levels of visibility, interpretability, and interactivity, marking a substantive scholarly progression.

V. CASE STUDY

To substantiate the practical applicability and efficacy of AI techniques in power systems, we present a case study on DSA. DSA plays a crucial role in assessing and managing the intricate challenges posed by RESs in modern power grids. It ensures the stability and reliability of the grid amidst the dynamic characteristics of these resources.

Additionally, to conduct a comparative analysis of the performance improvements attributable to the use of TL versus traditional methods, a robust yet simple domain adaptation approach, termed Adaptive Batch Normalization (AdaBN) is utilized. This feature-based TL technique significantly enhances the extensibility and generalizability of the DSA model under changing topologies, eliminating the need to train multiple models. This approach achieves a deep adaptation effect by modulating the statistics from the source domain to the target domain across all Batch Normalization layers within the model.

To evaluate the effectiveness of the proposed method, it is implemented on the IEEE 39-bus system. The system's layout includes 39 buses interconnected by 46 transmission lines, with 10 synchronous generators located strategically across different buses. It serves as the reference system in this study's simulations, as illustrated in Fig. 13. The development of the proposed method involves the integration of Digsilent Power Factory and Python for implementation and analysis.

In the process of generating the database for this study, the dataset is compiled by systematically sampling various operating conditions. This includes introducing random variations in loads within practical ranges, typically ranging from

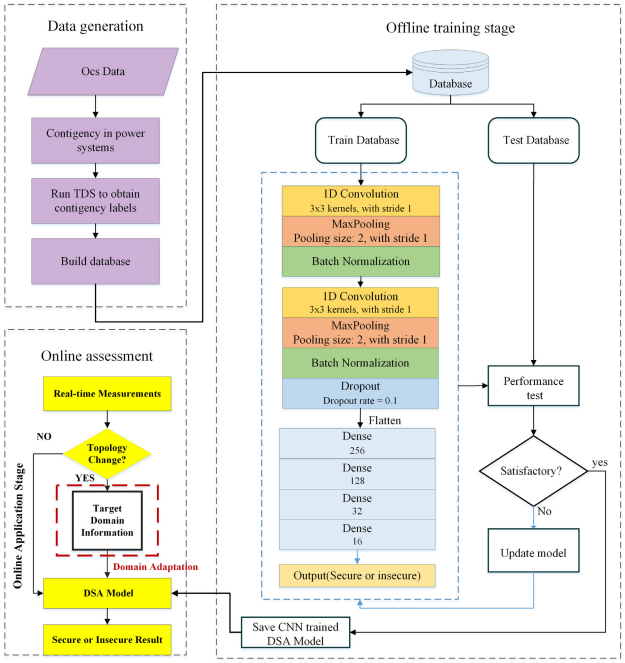


FIGURE 14. Flowchart of the proposed method.

TABLE 9. Comparison of the DSA performance for TL-based method and traditional method.

Metrics (%)	CBDAC [77]	SPORF [60]	RF [269]	Proposed TL-based approach
Accuracy	93.43	86.96	85.27	97.94
Precision	93.49	88.88	87.34	98.47
G-mean	92.32	85.02	83.21	97.12
F1-score	90.90	82.71	80.63	97.46

0.7 to 1.25 times the base load at each bus node. The resultant operating conditions encompass parameters such as active power generation (P_a), reactive power generation (P_r), active power of branches (L_a), reactive power of branches (L_r), generator bus voltage magnitudes (V), and their respective angles (θ), derived through optimal power flow calculations. Subsequently, each operational point undergoes time-domain simulation to determine its post-fault security status, categorized as secure or insecure based on criteria defined in the study. Specifically, if the rotor angle difference of at least two generators exceeds 180 degrees post-fault [268], the system is labeled as insecure; otherwise, it is considered secure.

The analysis incorporates eight distinct fault scenarios with interregional corridor trips (T1-T8), as detailed in Table 8, to evaluate system robustness under varying network configurations and fault conditions.

The flowchart of the presented method, as well as the architecture of the CNN-based assessment model utilized in this study, is represented in Fig. 14. The CNN incorporates 1D convolutional layers, max pooling, dropout layers, and batch normalization, leveraging these components to extract features, reduce dimensionality, stabilize training, and mitigate overfitting effectively.

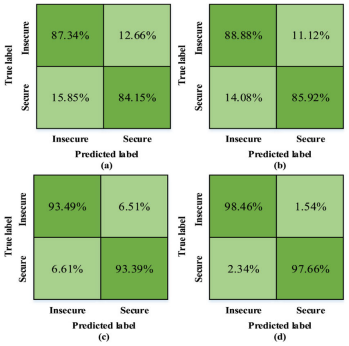


FIGURE 15. Confusion matrix of (a) RF [269] (b) SPORF [60] (c) CBDAC [77] (d) TL-based method.

During offline training, the CNN model is optimized to minimize discrepancies between predicted and actual states. To achieve effective model training, it is crucial to define a suitable loss function and employ an optimization algorithm. In this context, the model's predictions are evaluated against actual states using the cross-entropy (CE) function, which has demonstrated efficacy in classification tasks and ensures a robust measure of error. In this study, CE serves as the loss function, optimized using Adam's algorithm, a popular choice in DL due to its ability to efficiently minimize the loss function by iteratively updating model parameters. Adam's algorithm is noted for its rapid convergence and strong generalization performance, particularly beneficial in addressing DSA classification challenges. In the online application stage, real-time data collected by PMUs serves as input for the trained DSA model, previously optimized during offline training.

The model quickly determines the system's security status, prompting immediate corrective actions in case of identified insecurities.

A. EVALUATION OF MODEL PERFORMANCE

In this section, the performance efficiency of the proposed DSA model is evaluated using four indicators for the CNN-based DSA method (as presented in Fig. 14), and for a TL-based method to handle topology changes.

In real-world power systems, topology frequently changes because of control actions and maintenance activities, potentially making these topologies unfamiliar to the DSA model. Therefore, models often exhibit reduced accuracy when applied to different topologies, necessitating retraining and complicating online DSA efforts. Conversely, it is impractical to generate and label sufficient training data to train the DSA model for every possible topology.

By considering the system's topology as a domain, the domain from which labeled data is obtained and on which the model is trained is referred to as the source domain. Conversely, the domain where unlabeled data is collected, and the model is tested is referred to as the target domain.

It is crucial for a classifier which is trained on samples from a specific topology to maintain high accuracy when applied to other topologies and effectively transfer knowledge from the target domain to the source domain. This scenario frames the issue as a domain adaptation problem.

In this section, we assess the proposed model's performance on unfamiliar topologies. Eight DSA classification models were trained, each corresponding to a different database. Then, domain adaptation was utilized to transfer each of the seven topologies to the remaining one, followed by an evaluation of the performance. Table 9 presents a comparison of the presented method with traditional techniques (without domain adaptation) using various evaluation metrics. The results indicate that the presented method, which incorporates TL, outperforms traditional methods by achieving an accuracy of 97.94% for unseen topologies. In comparison, the highest accuracy obtained by traditional methods is 93.43%. These findings demonstrate that domain matching through TL is an effective solution for addressing topology changes in machine learning-based DSA methods.

The confusion matrices corresponding to the methods in Table 9 are presented in Figure 8a-d. As illustrated in Fig. 8d, the proposed TL-based approach has performed very well in identifying insecure operating points across previously unseen topologies. A detection rate of 98.46% is achieved by employing this domain adaptation-based approach, which is comparatively higher than traditional approaches. Moreover, the TL model has resulted in fewer False positives, which are insecure operating conditions falsely predicted as secure [270], as shown in the confusion matrix of the proposed method. This outcome is crucial for ensuring the safe operation of power systems.

VI. CONCLUSION

This study presented a comprehensive literature review, incorporating the latest research from the past decade, focusing on 2020 to the present. Detailed examinations of four primary AI applications in power systems—load forecasting, security assessment, voltage stability assessment, and LS—were provided alongside exploring four additional applications (fault detection and location, PQD analysis, state estimation, and FDIA detection and localization). The discussion mentioned various features of AI-based solutions and also delved into the challenges associated with their practical implementation in power systems.

Our analysis of various papers demonstrated that while these studies yielded promising results with high accuracies, they still struggled with challenges related to the practical implementation of AI in real-world power systems. These challenges stemmed from the dynamic nature of power systems (e.g., Variations in network topology resulting from maintenance or control actions, fluctuations in operational conditions and load levels) and various data-oriented issues (e.g., Imbalanced or insufficient datasets for effective model training, data scarcity for rare fault scenarios, inadequate representation of system dynamics, discrepancies between simulation models and real-world systems). Importantly, it was evident that more than relying on ML models was needed to tackle these complex challenges. Relying solely on conventional ML models is insufficient for addressing these multifaceted challenges.

In response to these identified limitations, our study introduced two powerful tools, TL and digital twin technology, that demonstrated substantial potential in overcoming these barriers. TL addresses critical data-related challenges by leveraging knowledge gained from one context to tackle issues in another context or domain, effectively mitigating data scarcity, enhancing generalizability, and improving prediction accuracy. Digital twin technology addresses physical, real-world limitations by creating accurate virtual replicas of physical systems. Real-time monitoring, predictive analytics, and scenario evaluations can be conducted without the limitations of physical experimentation by employing digital twin technology in a power system.

The key takeaway of this study is that integrating TL and digital twin technology significantly enhances prediction accuracy, operational reliability, and decision-making quality. Furthermore, this combined approach enables extensive scenario testing and evaluations on virtual models. This capability is virtually impossible in physical environments due to practical constraints (e.g., safety concerns, high costs, limited accessibility, and time constraints). This integrated approach is particularly interesting as it systematically addresses significant barriers traditionally faced in power system management. Overcoming data and physical constraints simultaneously provides stakeholders with robust, flexible, and insightful tools for advancing sustainable, efficient, and resilient power system operations.

As a result, our findings underscore the imperative for future research to shift its focus toward the seamless integration of digital twin and AI technologies. This integrative approach has advanced the field, providing more effective solutions for navigating power systems' intricate and dynamic landscape.

In order to advance TL methodologies, future research should prioritize the exploration of more efficient strategies to mitigate domain disparities and prevent negative knowledge transfer. This entails developing adaptive techniques that can dynamically align source and target domains, thereby reducing domain mismatch risks. Additionally, there is a critical need to innovate in the augmentation and selection of pre-training datasets, ensuring they are tailored to diverse application domains to enhance model robustness and generalization capabilities. By addressing these challenges, future studies can significantly bolster the efficacy and applicability of TL across varied real-world scenarios.

Regarding the digital twin technology, it is imperative to prioritize the establishment of global standards to facilitate the wide-scale implementation of digital twin and foster robust collaboration between the industry and community

of science. Additionally, research efforts should concentrate on resolving the unique complexities of power systems to advance the development of more precise simulations and predictions. This approach will drive the successful integration of digital twin in the power sector, ultimately shaping a more optimized, reliable, and resilient energy landscape.

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